Potential applications of negative radiation pressure?

e.g. 2WQC, medical, astro?

Jarek Duda, rec, arXiv:2409.15399 , 2WQC

Radiation pressure: $\vec{p} = \frac{\langle \vec{E}\times\vec{H}\rangle}{c} = \frac{I_e}{c}$ (Poynting)

photon **emission**, **toward** surface: **positive/push**

Negative radiation pressure: **outward** surface, e.g. **reverse**: **T-symmetry**, $\vec{H} \to -\vec{H}$

nature.com/articles/s41598-022-10699-7 **positive**/**negative** in metamaterials,

gain medium, pulling kinks … **EM pulling energy from resonators like atoms?**

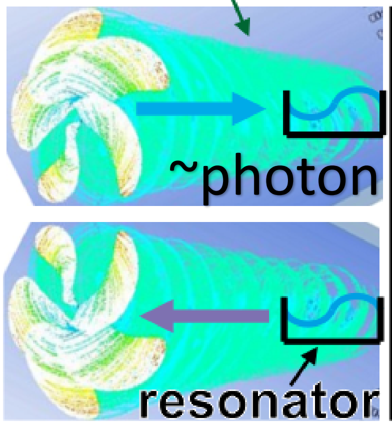
T-symmetry switches between **absorption** and **stimulated emission** equations

EM ~ hydro

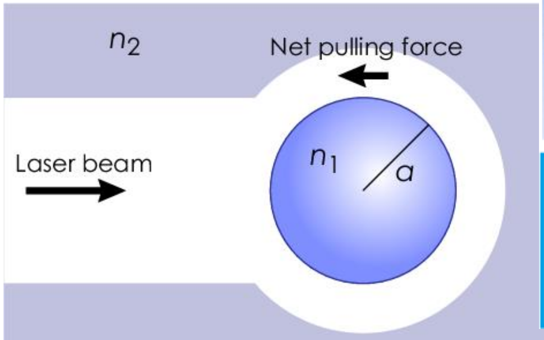
Theory	Gauge fields	Circulation	Gauge condition	Matter field
Electro-dynamics	$\varphi \quad ,\vec{A}$ four-potential	$\vec{B} = \vec{\nabla}\times\vec{A}$ magnetic f.	$\vec{\nabla}\cdot\vec{A} + \frac{1}{c^2}\frac{\partial\varphi}{\partial t} = 0$	$\vec{E}_e = -\frac{\partial\vec{A}}{\partial t} - \vec{\nabla}\varphi$
Hydro-dynamics	$\chi = v^2/2, \vec{v}$ flow velocity	$\vec{\omega} = \vec{\nabla}\times\vec{v}$ vorticity	$\vec{\nabla}\cdot\vec{v} + \frac{1}{c_s^2}\frac{\partial\chi}{\partial t} = 0$	$\vec{E}_h = -\frac{\partial\vec{v}}{\partial t} - \vec{\nabla}\chi$

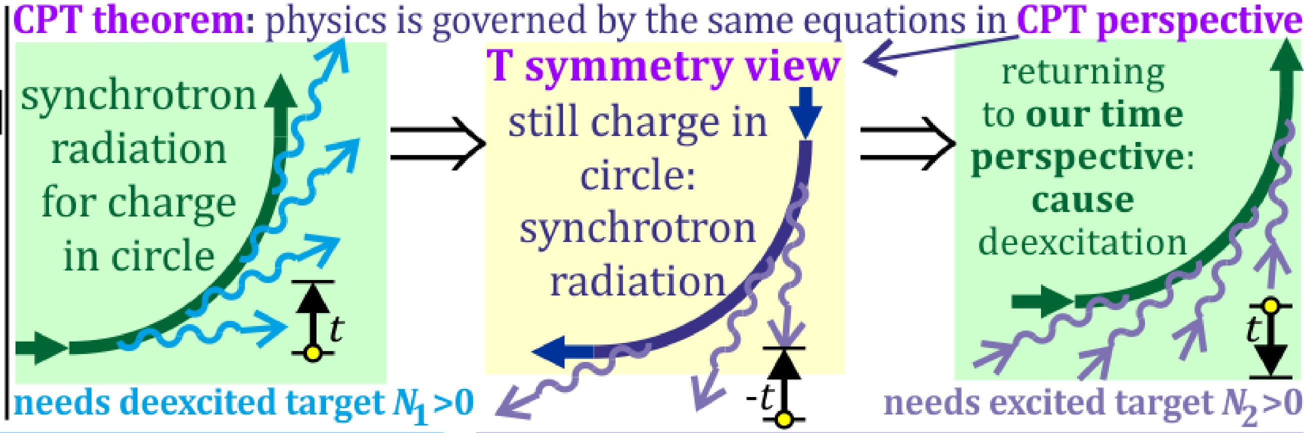
EM Barnett effect: uncharged magnetized by spinning $B \propto \omega$

$E, \vec{p}, \vec{L}$ like **photon**:



Optical pulling:





$$\frac{\partial N_2^{\text{excited}}}{\partial t} = -\frac{\partial N_1^{\text{ground}}}{\partial t} = B_{12}\,\rho(\nu)\,N_1$$

usually ~N — absorption

T

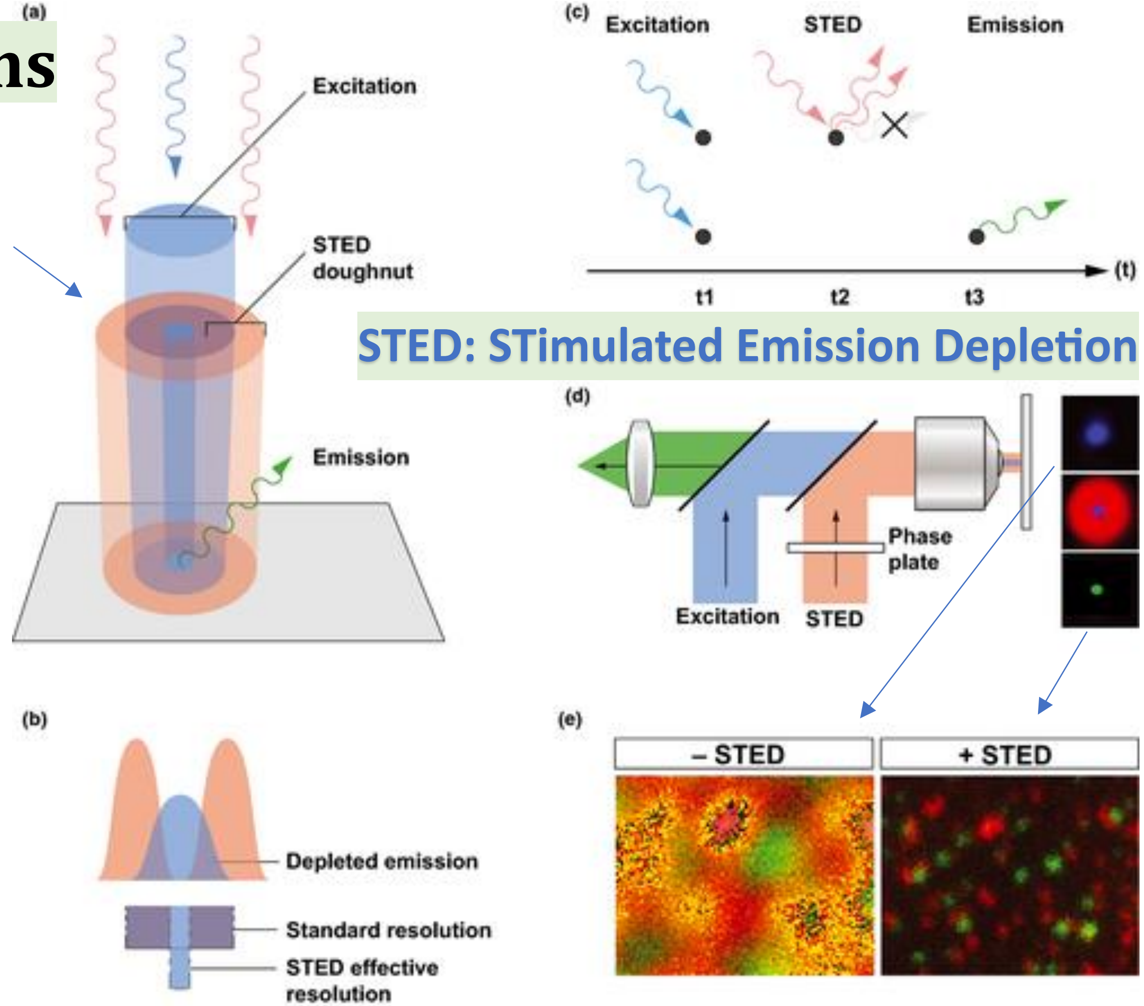
$$\frac{\partial N_2^{\text{excited}}}{\partial t} = -\frac{\partial N_1^{\text{ground}}}{\partial t} = -B_{21}\,\rho(\nu)\,N_2$$

usually ~0 — stimulated emission

Laser causes **absorption** $\xleftrightarrow{\text{CPT}}$ **stimulated emission** e.g. in Rabi cycle – let's use the latter

Stimulated emission applications with negative radiation pressure

Like Rabi cycle, STED microscope



STED: STimulated Emission Depletion

apply/test macro CPT symmetry

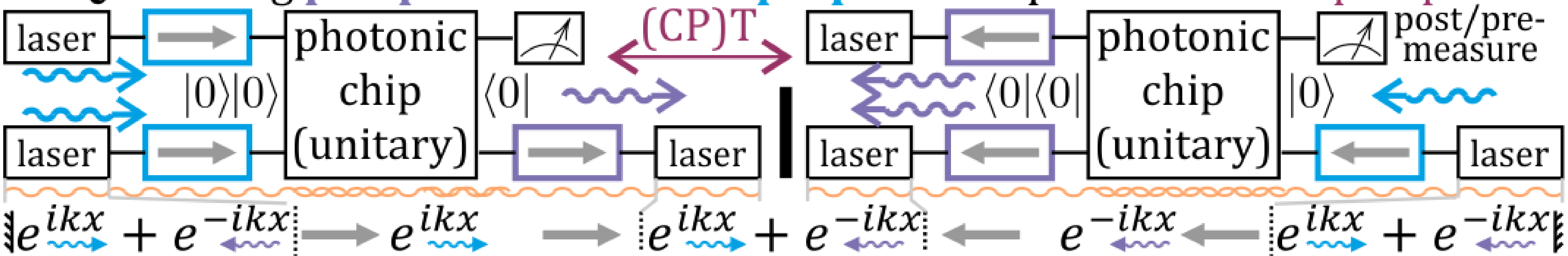
CT scan of emission coefficient
as CPT analog of absorption CT,
radiotherapy to starve tumor,

astro: **CPT(black hole) only emits**?
surprising neg. radiation objects?

2WQC: two-way quantum computers

S-matrix: $\langle\psi_f|U|\psi_i\rangle \xleftrightarrow{\text{CPT}} \langle\psi_i|U^\dagger|\psi_f\rangle$

2WQC: adding **postparation** as state **preparation** process in CPT perspective



Couple laser resonators around photonic chip (directing by optical isolators?)

[opg.optica.org/ oe-20-9-9501](opg.optica.org/oe-20-9-9501) **two → [three tested 2024](#)**

Superradiance using negative radiation pressure

forward bias diode (LED): tendency to emit + T-symm.:

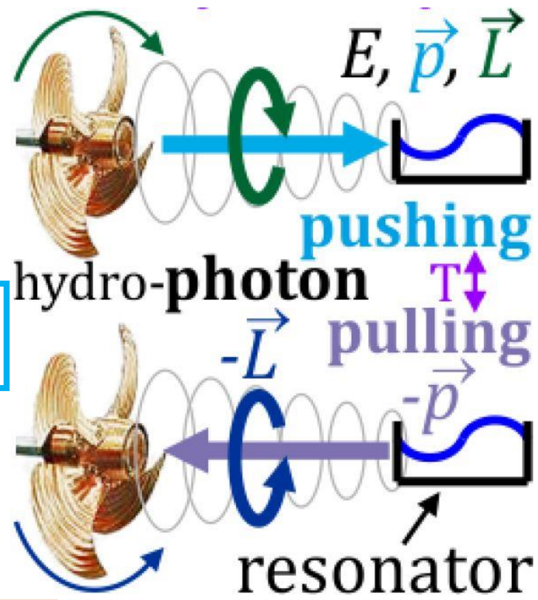
reverse bias diode: tendency to absorb e.g. in [detectors](#)

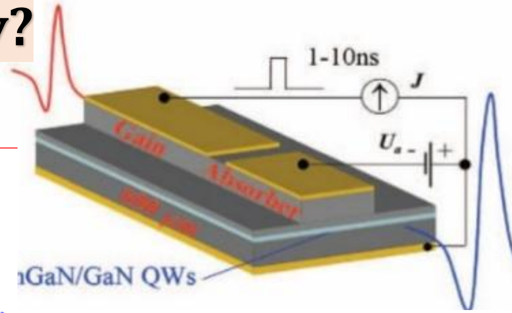
[S-matrix](#) $\langle\psi_f|U|\psi_i\rangle$: both control photon exchange

Causing absorption/stimulated emission on target

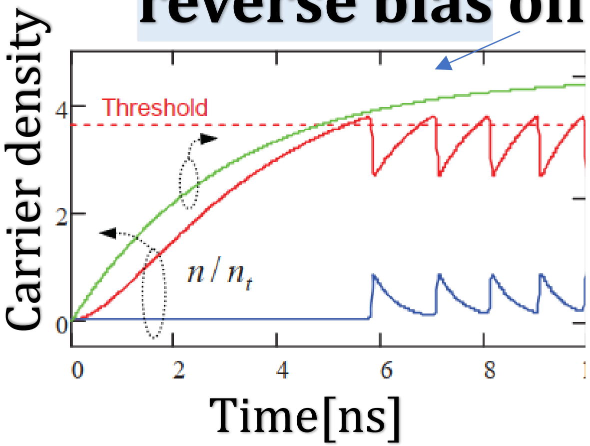
Reverse biased actively helped with emission:

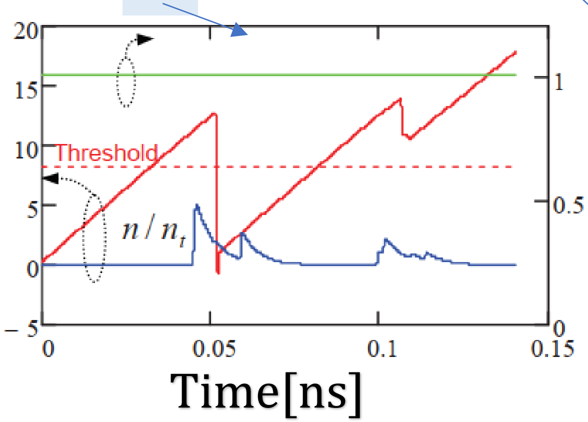
more than just absorber, getting superradiance

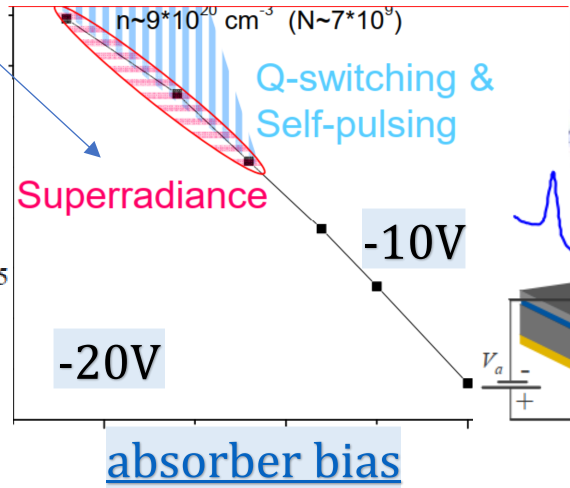


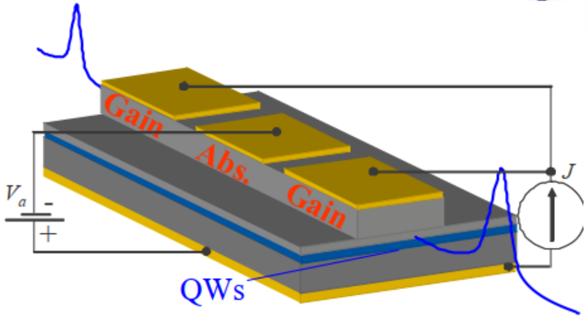


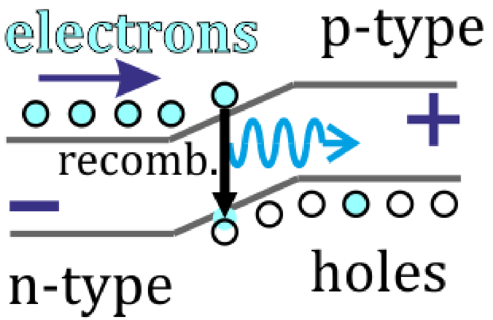
reverse bias off **on**



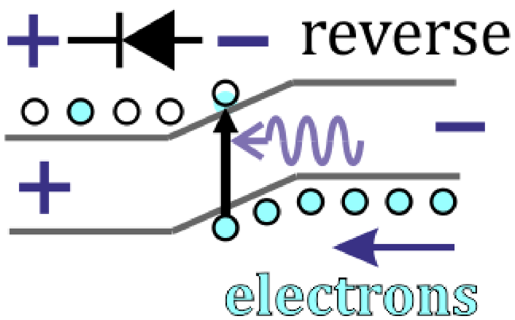


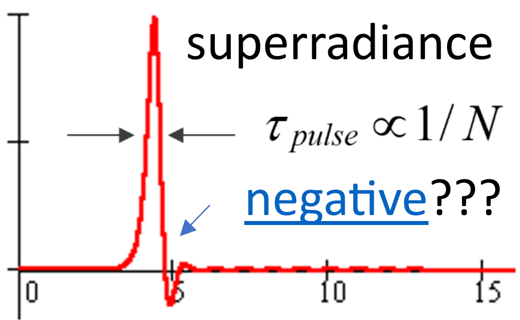






light diode gets
tendency to absorb
for **reverse bias**:
electron gradient





Physics should be governed by the same equations in CPT symmetry perspective

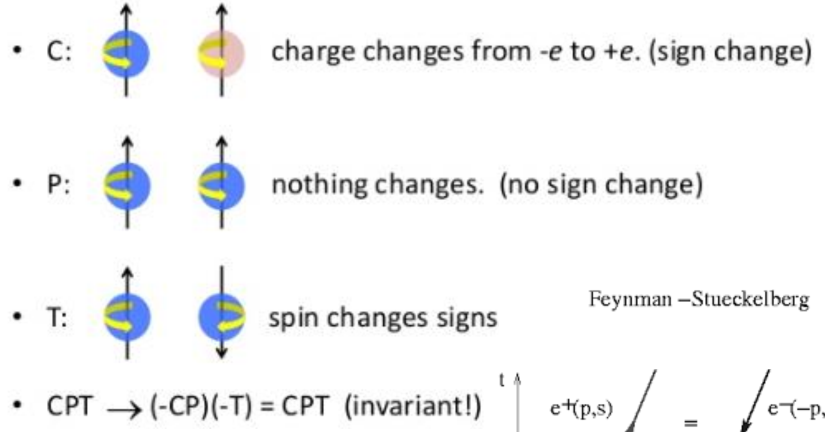
C – charge conjugation, P – parity, **T - time**

"The CPT theorem says that [CPT symmetry](#) holds for all physical phenomena (...)" ([link](#))

any **Lorentz invariant** local **quantum field theory** with a **Hermitian Hamiltonian must have CPT symmetry**"

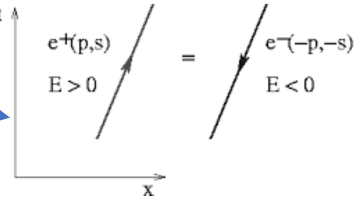
"**CPT Violation Implies Violation of Lorentz Invariance**"

CPT for Electron



Feynman–Stueckelberg: "antiparticles travel backward in time"

Feynman –Stueckelberg



Many **microscopic confirmations**: "Data Tables for Lorentz and CPT Violation"

Macroscopic tests? Applications like 2WQC? EM T-sym: Feynman-Wheeler absorber theory

CPT symmetry in equations governing physics

Can be violated in solutions e.g. 2nd law of thermodynamics

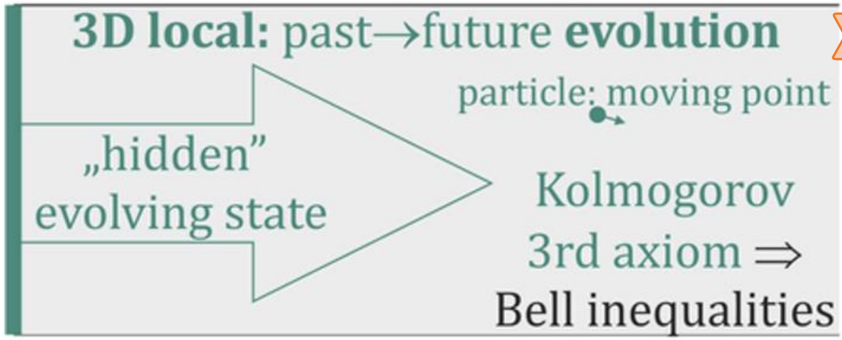
Big Bang as 'the rock'? Everything localized: low entropy

presentism

~~CPT~~ **presentism**

directional evolution

like wave propagation

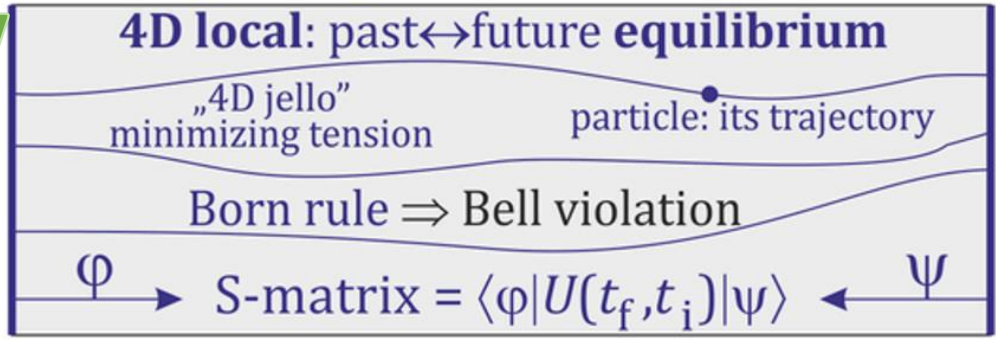


X V

CPT eternalism, block universe GR, QFT

symmetric boundary conditions

static e.g. Ising model, **4D spacetime**



TSVF, $G_{\mu\nu} \propto T_{\mu\nu}$

eternalism

2WQC - exploit **S-matrix** formulation: $\langle\psi_f|U|\psi_i\rangle \overset{CPT}{\longleftrightarrow} \langle\psi_i|U^\dagger|\psi_f\rangle$

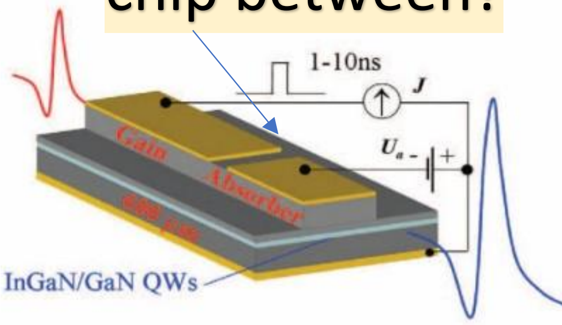
Photonic: emission forward bias + absorption reverse bias

symmetric two-way **2WQC**: preparation $|0\rangle$ + postparation $\langle 0|$

$\langle 0|$ = CPT(process used for $|0\rangle$), e.g. reversing used EM impulses

solving postBQP ∋ NP, better error resistance and correction

chip between?



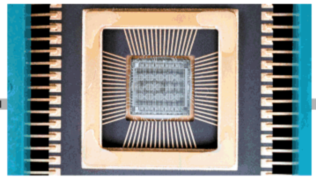
Push&pull for better flow control

electrons battery as "pump"

+ −

electronic chip

pull push

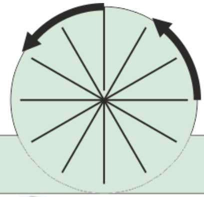


Hydrodynamics

fluid pump

negative pressure

positive pressure

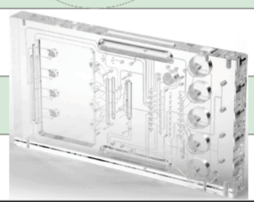


microfluidic chip

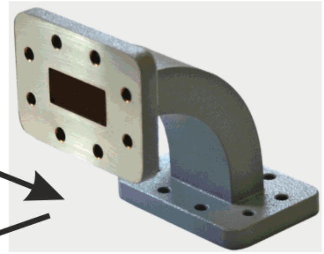
superfluid

QC?

pull push



microwave waveguide quant. chip?



ring laser

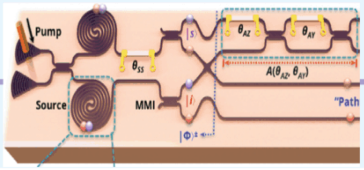
(radiation)

negative pressure

positive pressure

photonic chip?

CPT(pump) pump



time/CPT symmetry

photons

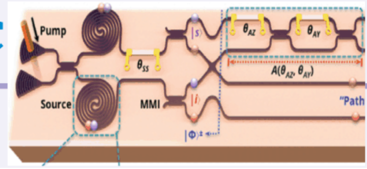
(radiation)

positive pressure

negative pressure

photonic chip?

pump CPT(pump)



EM field (photons?) nearly the same equations as **superfluid** (mechanical vibrational qubits?)

CPT(process used for state preparation) to influence the final state

setting	Gauge fields	Circulation	Gauge condition	Matter field
Electro-dynamics	φ , $\vec{A}$ four-potential	$\vec{B}=\vec{\nabla}\times\vec{A}$ magnetic f.	$\vec{\nabla}\cdot\vec{A}+\frac{1}{c^2}\frac{\partial\varphi}{\partial t}=0$	$\vec{E}_e=-\frac{\partial\vec{A}}{\partial t}-\vec{\nabla}\varphi$
Hydro-dynamics	$\chi=v^2/2,\vec{v}$ flow velocity	$\vec{\omega}=\vec{\nabla}\times\vec{v}$ vorticity	$\vec{\nabla}\cdot\vec{v}+\frac{1}{c_s^2}\frac{\partial\chi}{\partial t}=0$	$\vec{E}_h=-\frac{\partial\vec{v}}{\partial t}-\vec{\nabla}\chi$

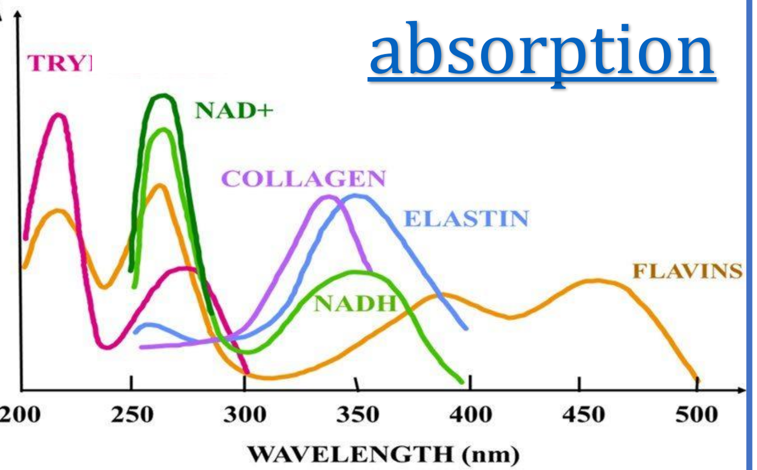
Medical applications?

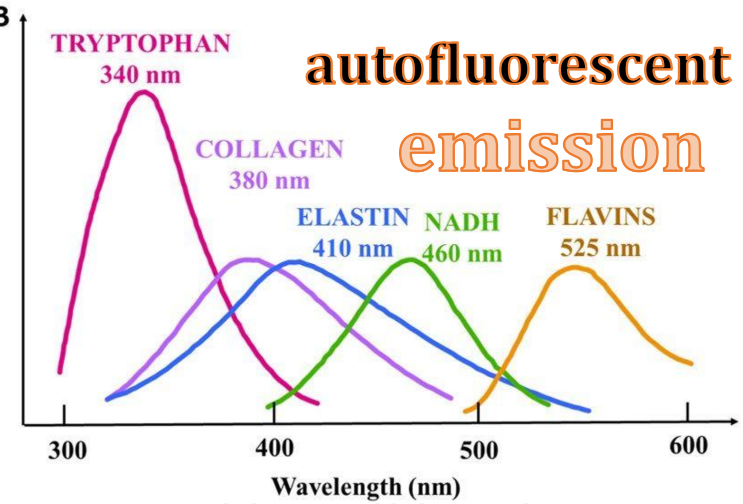
[arXiv:2409.15399](https://arxiv.org/abs/2409.15399)

e.g. mapping emission coefficient?

e.g. in human body

mobile if on diodes



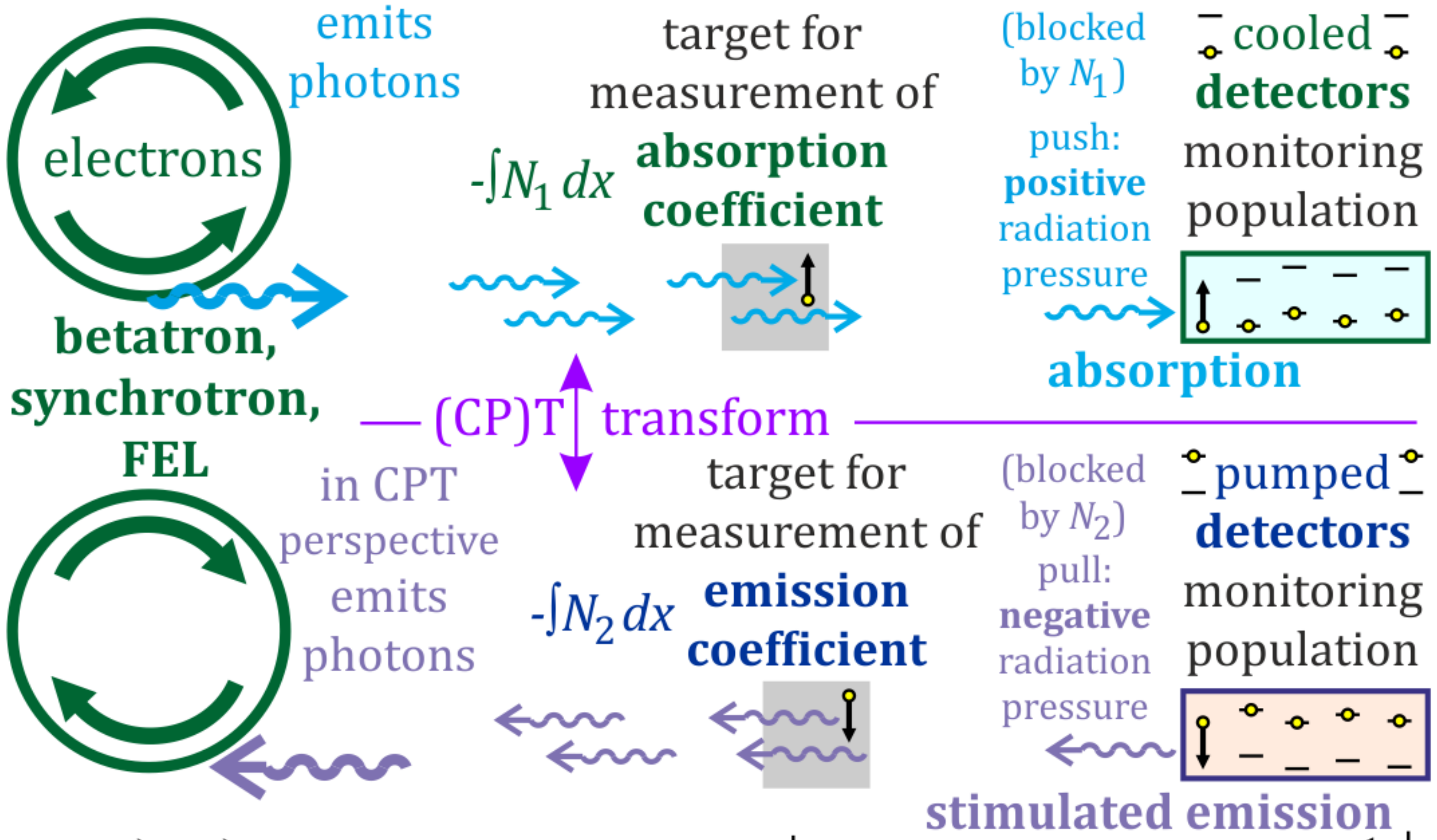


powered by cell biochemistry

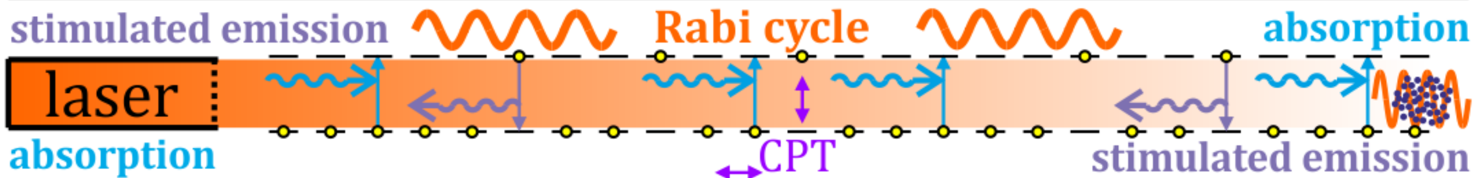
Harmless (non-ionizing) color RTG/CT? (video?)

CT scan of emission coeffcient for 3D map of e.g. tryptophan ~340nm, NADH ~460nm, flavins ~525nm emission

should have **much better transparency** as usually $N_2 << N_1$



$\vec{p} = \langle \vec{E} \times \vec{H} \rangle / c$ **radiation pressure** - positive (toward surface) or negative (outward)



$$\frac{\partial N_2^{\text{excited}}}{\partial t} = -\frac{\partial N_1^{\text{ground}}}{\partial t} = B_{12}\,\rho(\nu)\,N_1$$

usually ~N — absorption

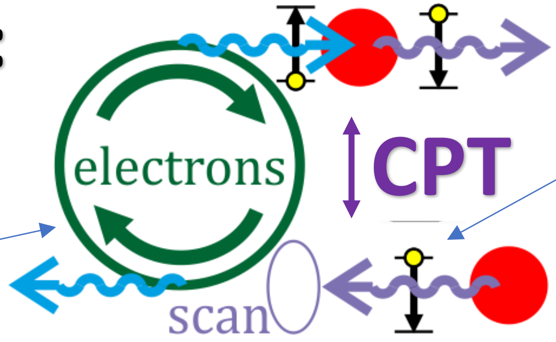
$$\frac{\partial N_2^{\text{excited}}}{\partial t} = -\frac{\partial N_1^{\text{ground}}}{\partial t} = -B_{21}\,\rho(\nu)\,N_2$$

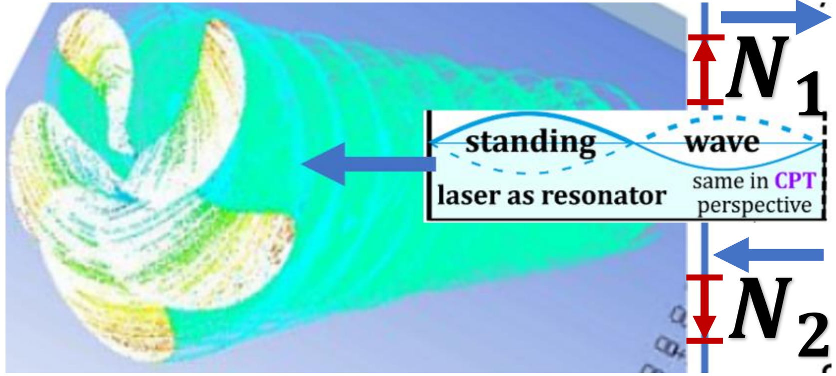
usually ~0 — stimulated emission

Just cause deexcitation:

STED microscopy …

without photobleaching?



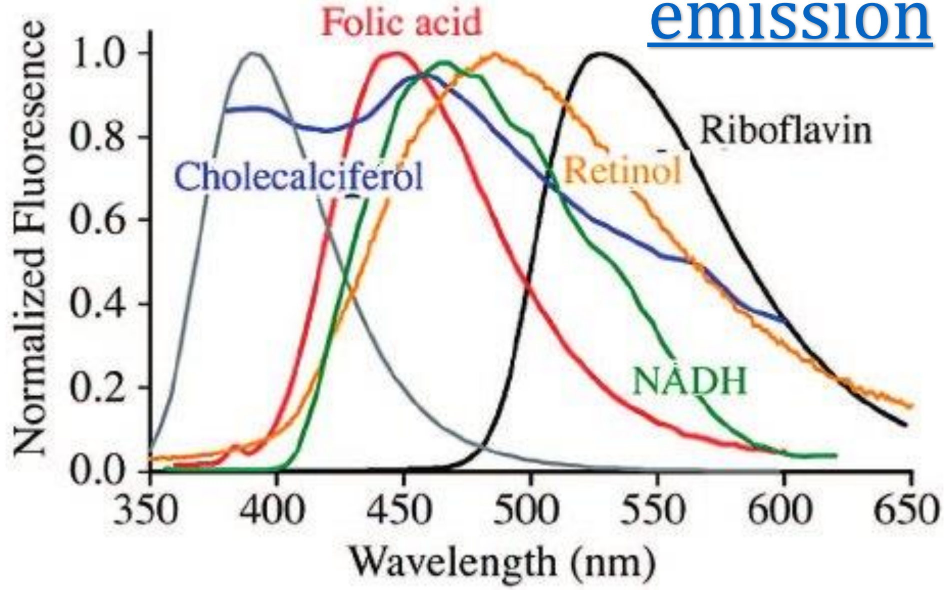


separation (positive/negative pressure)

transparency: blocked by N_2 not N_1

2WQC, reversed photolithography, chemistry, nuclear, neuro, medicine?

emission



e.g. degrade NADH to starve cancer?

precise, non-ionizing

neutral for intermediate tissues

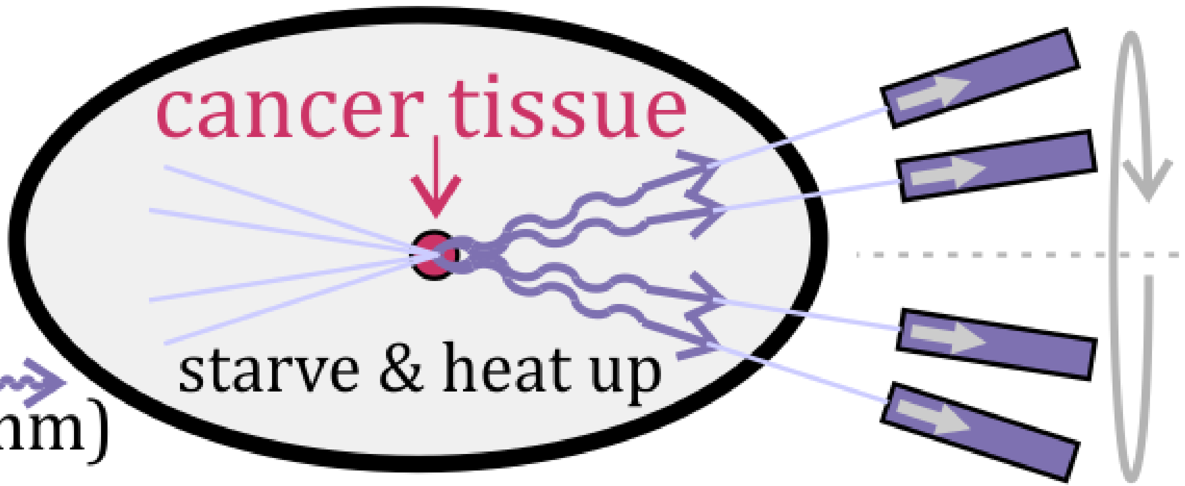
Metabolic issues? (toxic) singlet oxygen ~1280nm IR emission? …

Backward ASE radiotheraphy?

amplify chosen photon emission by intersecting **backward beams**

e.g. stimulate NADH degradation to starve, release energy heating

$NADH \rightarrow NAD^+ + H^+ + 2e^- + \gamma$ (~460nm)



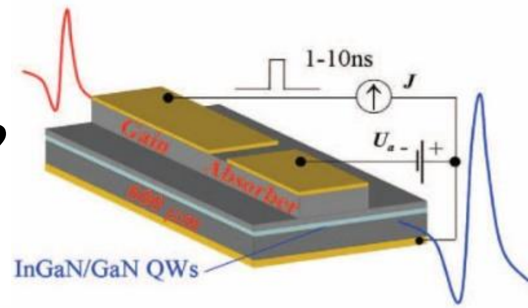
Are there **cosmic sources** of **negative radiation pressure**?

Astronomy: now focused on absorption - of positive radiation pressure

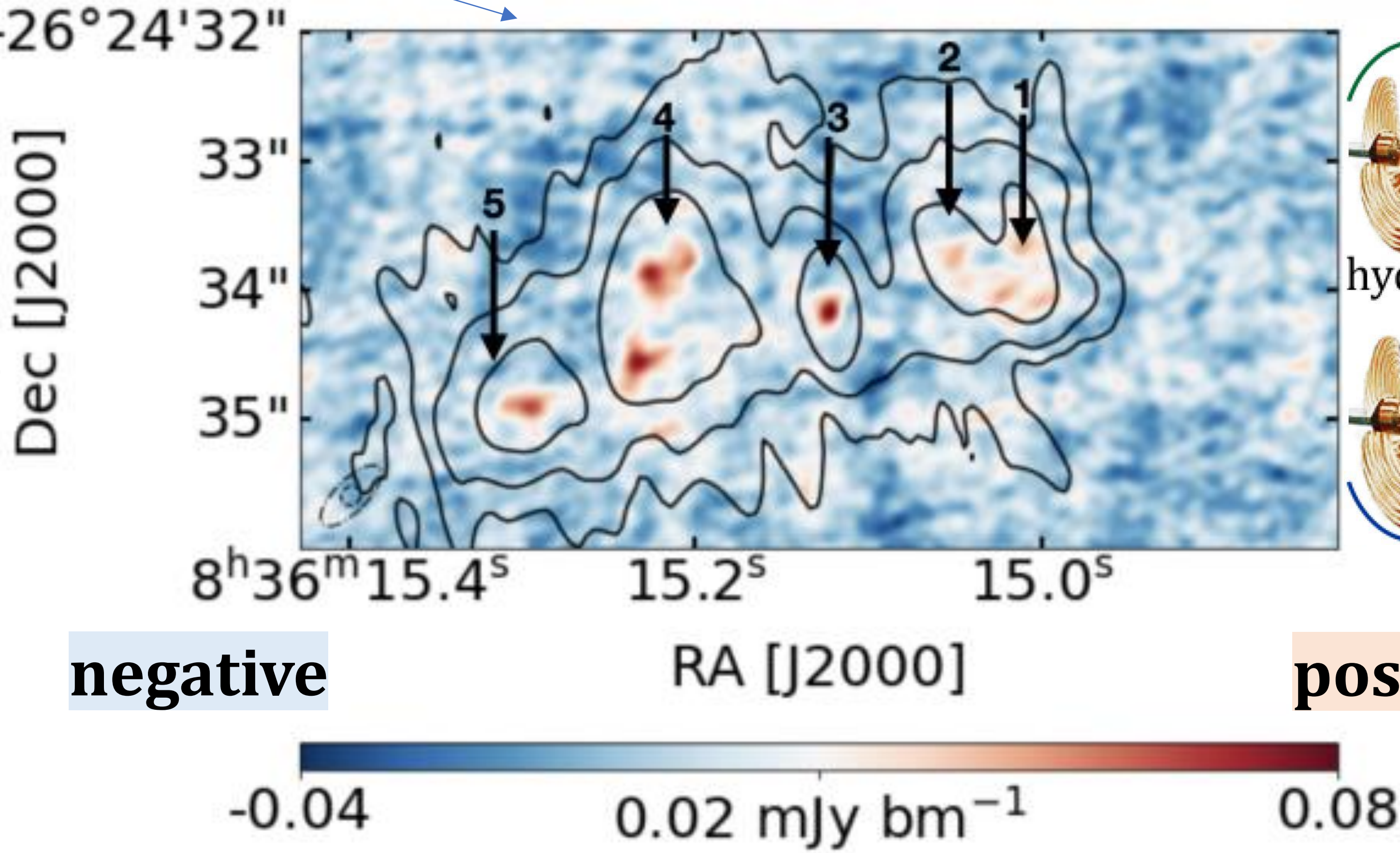
Negative: telescope emits more than absorbs?

Reverse-bias-diode-like sources? Pulsar? Black hole?

Observed e.g. in radio flux maps – really artifacts?



[arXiv:2107.02695 Astrophysical Journal](arXiv:2107.02695)



negative **positive**

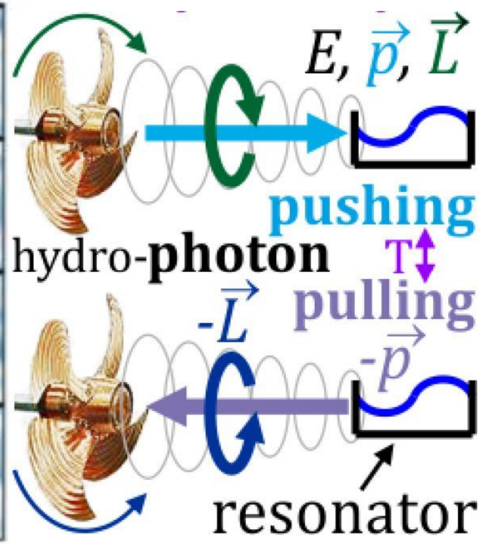


Figure 1. 33 GHz continuum map of Henize 2-10. The contours are the 5 GHz map from VLA/AJ314 to show radio knots as originally classified by Kobulnicky & Johnson (1999). The 5 GHz contours are at 5, 10, and 22σ. The beams are shown in the lower left.

$\langle \psi_f | U | \psi_i \rangle$: **white hole** perfect emiter $|\psi_i\rangle \xleftrightarrow{\text{CPT}}$ **black hole** perfect absorber $\langle \psi_f|$

From CPT perspective, our black hole is white hole: very active

E.g. CPT(white hole heating around) = black hole cooling around?

Black hole information paradox: destroyed in BH evaporation???

Solution: white hole exchanges information by positive radiation pressure, hence (CPT) symmetrically black hole by negative radiation pressure

$$\frac{\partial N_2^{\text{excited}}}{\partial t} = -\frac{\partial N_1^{\text{ground}}}{\partial t} = B_{12}\,\rho(\nu)\,N_1$$

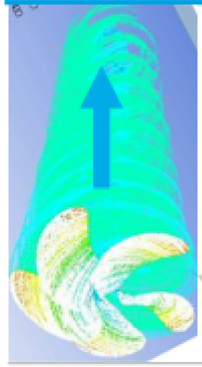
usually $\sim N$

absorption

$$\frac{\partial N_2^{\text{excited}}}{\partial t} = -\frac{\partial N_1^{\text{ground}}}{\partial t} = -B_{21}\,\rho(\nu)\,N_2$$

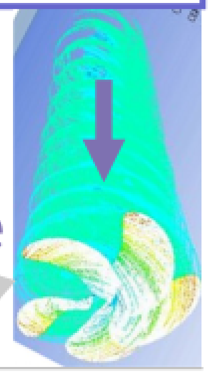
usually ~ 0

stimulated emission



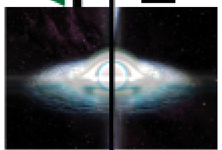
push, **emit**, **heat**, acting with **absorption** equation, positive radiation pressure, forms **jets going out**

pull, **absorb**, **cool**, acting with **stimuated emission**, negative radiation pressure forms **jets going in**



push
blow

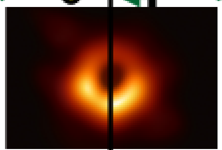
horizon



time

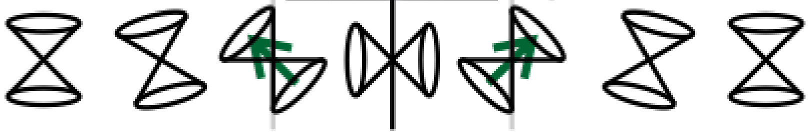
radiation pressure

$$\vec{p} = \langle \vec{E} \times \vec{H} \rangle / c = I/c$$



pull
suck

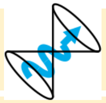
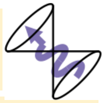
lightcones



white hole **T/CPT symmetry** **black hole**

arXiv:2509.10615 : “SMBHs are **expected** to be surrounded by progressively hotter gas (…) **Surprisingly** (…) Sagittarius A* (…) **seems to have no currently active jet or wind**.”

?: Intermediate mass BH $10^2 - 10^5\ M_\odot$: very low luminosity

CPT(WH  **) = BH** 

$\frac{\partial E}{\partial t}$ **CPT(heating) = cooling**

$\propto N_1 \gg \propto N_2$

“surprising”, “dust”?

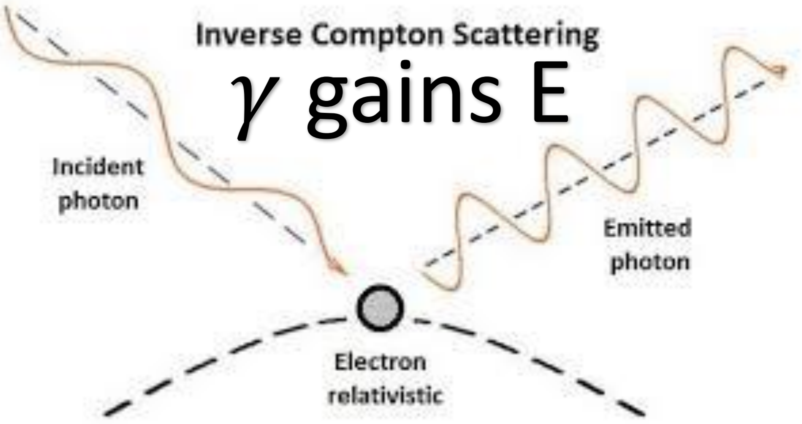
BH is WH in CPT view

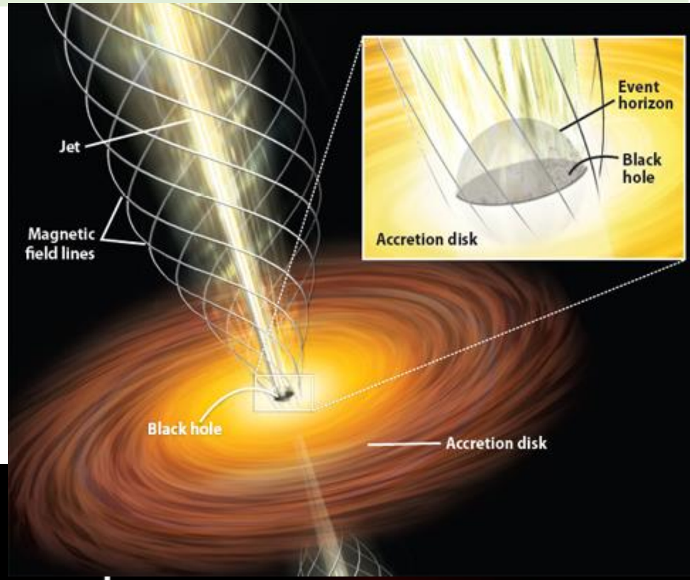
CPT violation otherwise?

BH simulations: **electron temperature << of ions! E.g. 160x**

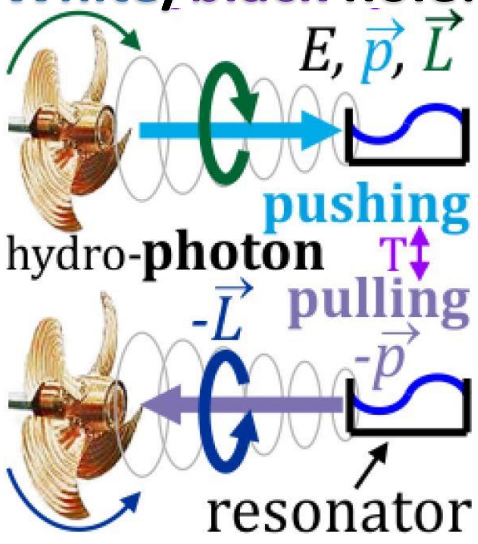
MAD, **SANE model**: "Near $R_{high} \approx 160$, the emitting region shifts significantly toward **higher latitudes** and **jet-disk boundary**” **cooling**: **inverse Compton** + **pull/stimulated emis.**?

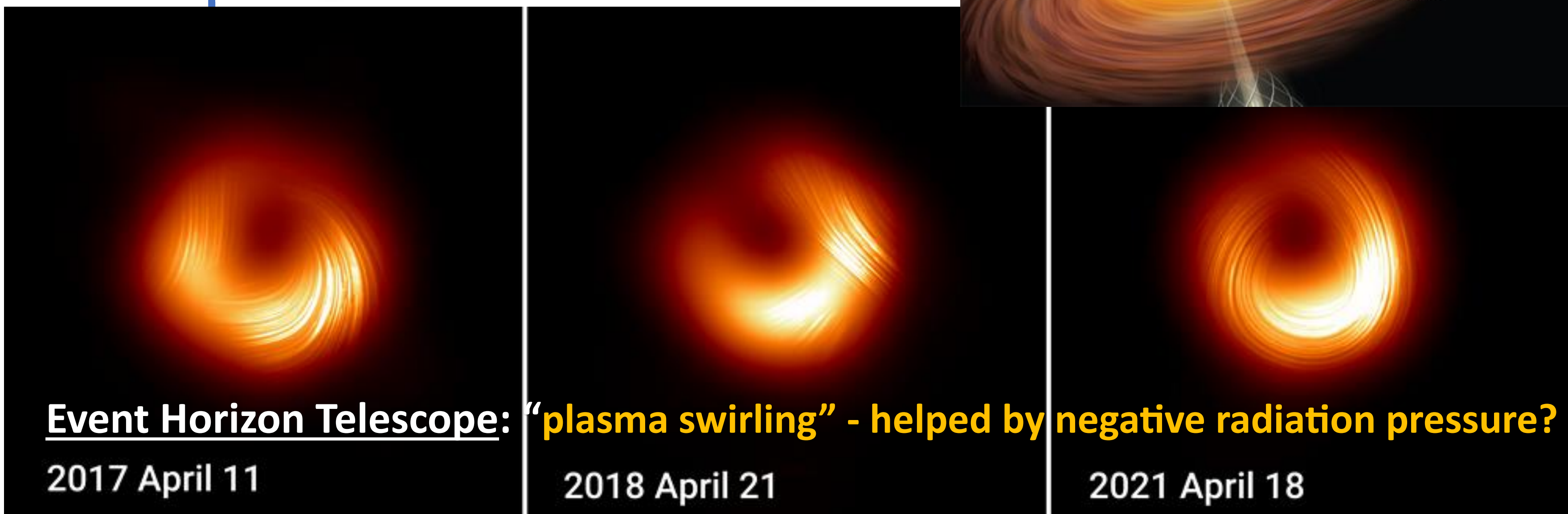
“Tremendous growth black hole”? by **pulling gravit. + EM**? Hawking?





White/black hole:





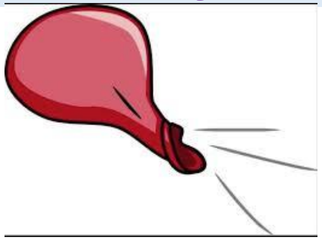
Negative-temperature pressure in black holes, Norte, EPL 2024

BH negative temperature? [1,2,3,4,5]

“**negative T yield negative pressures**”

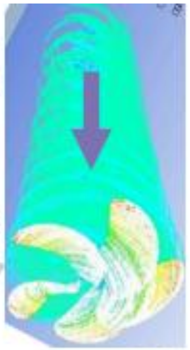
$T > 0$ matter: tendency to deexcite, radiating **positive** radiation pressure.

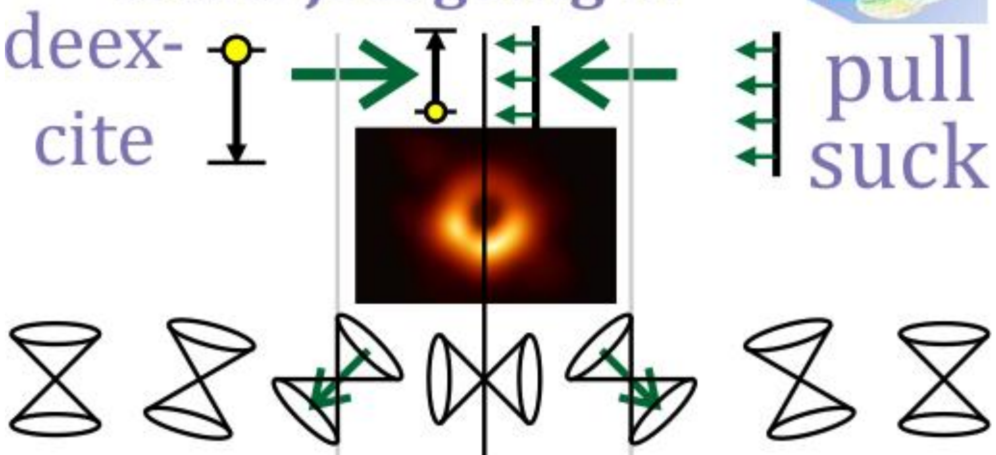
$T < 0$ matter: **tendency to excite** by **negative pressure**

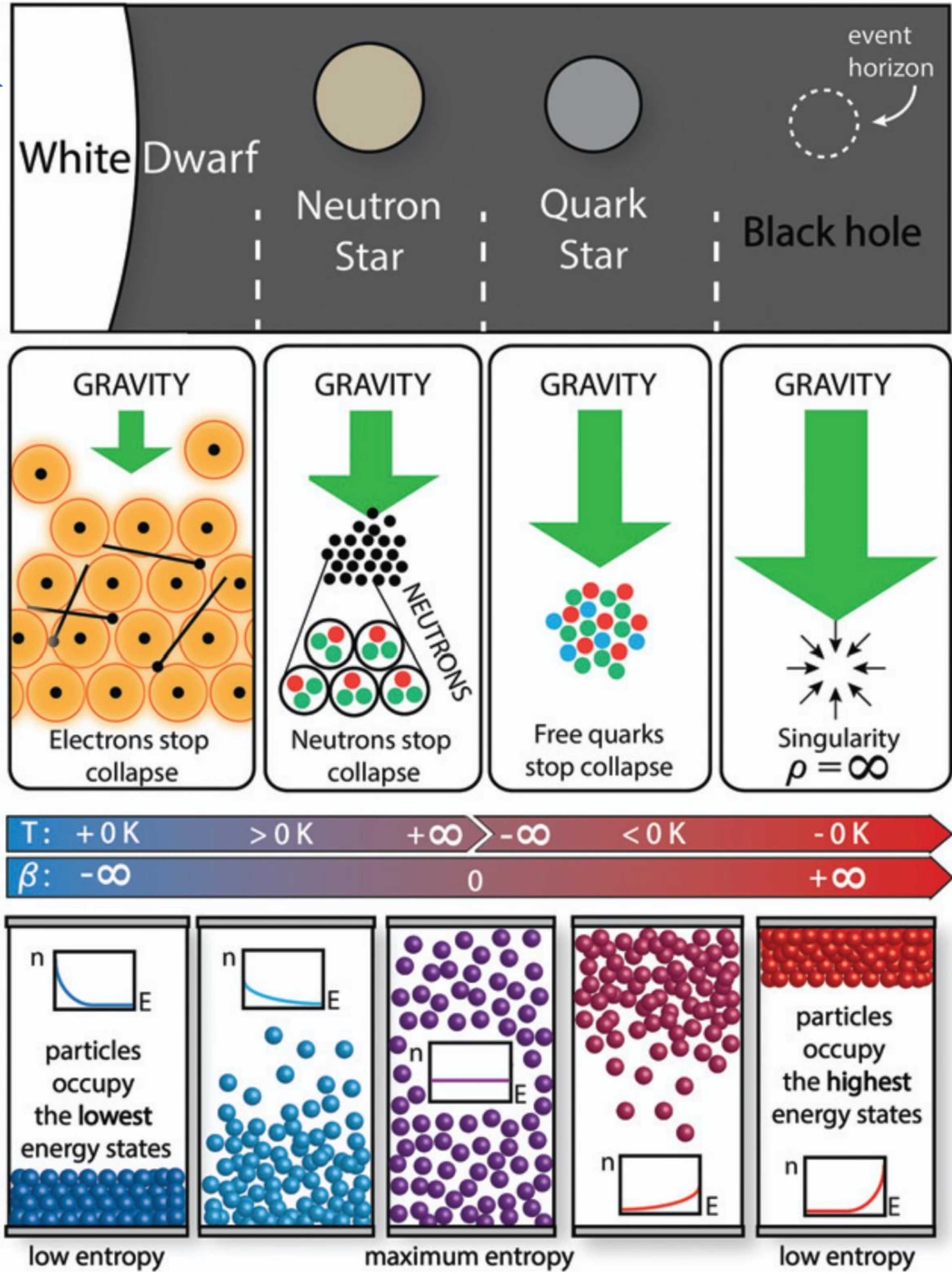


$t \leftrightarrow -t$ time

pull, **absorb**, **cool**, acting with **stimuated emission**, negative radiation pressure forms **jets going in**







Universe no longer accelerating? DESI: dark energy weakens?, revisit, Sabine: younger dimmer?

History of (4D block) Universe? (CPT) symmetry? Cyclic?

Big Bang

us

entropy time

heat death age
~equilibrium

thermodynamically as **Big Bang**?

γ γ ?

cyclic?
Big Bounce

2nd law of thermodynamics due to BB?

anchor of our reason → result relation chains

2nd law

cyclic?
entropy gradient?

Differences between **evolutions**: **forward Big Bang** vs **backward Big Crunch**?

spontaneous **emission**

coupled by photon practically **random**
two electrons in **4D Feynman diagram**

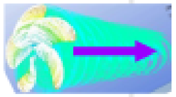
spontaneous absorption?

Asymmetry because of more **absorbers** in the **future** than **emitters** in the **past**?

stimulated: **synchrotron** mainly **emit/** **lose energy**

e e e

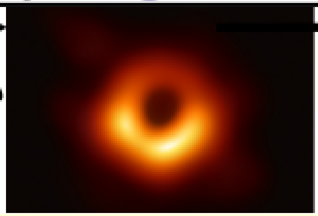
time



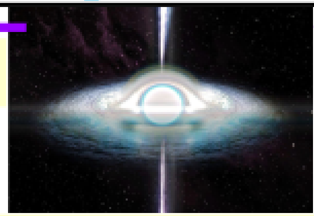
Mainly **negative** **radiation pressure** **positive** acting

synchrotron for us mainly **absorb/** **gain energy**?

black hole (forward)



? switched from T/CPT perspective ?



white hole??? (backward)

As white holes would **actively emit/push**, **shouldn't black holes** **absorb/pull**?

What **spectrum** should we expect from (lone) **white holes**? Black body radiation?

"**push through earth**" ... maybe "**pull by negative radiation pressure**"?

ANITA-I, III, IV, Fluorescence Detector of the Pierre Auger Observatory

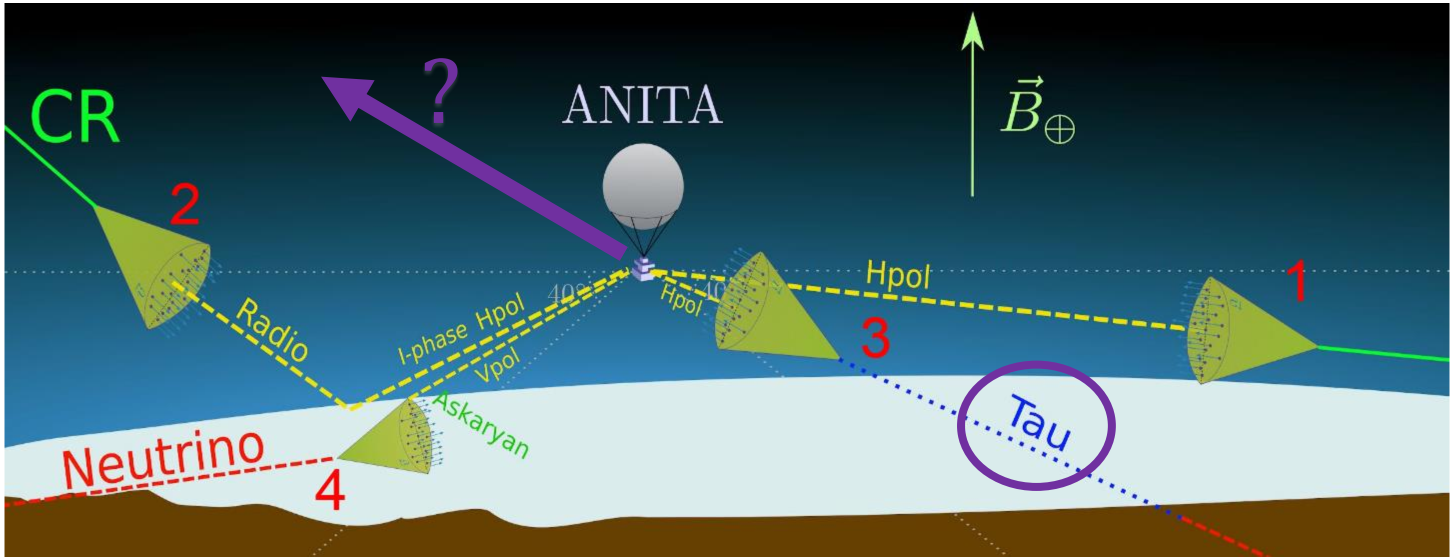
SM, ν_τ ($> 10^{18}$eV)

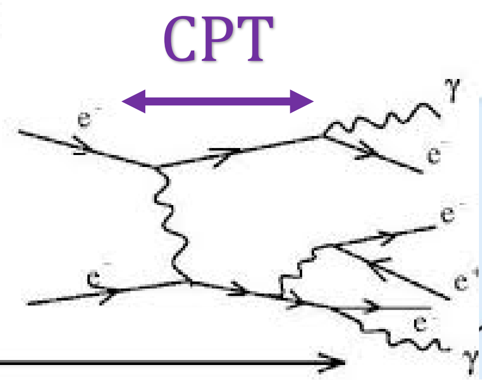
excluded

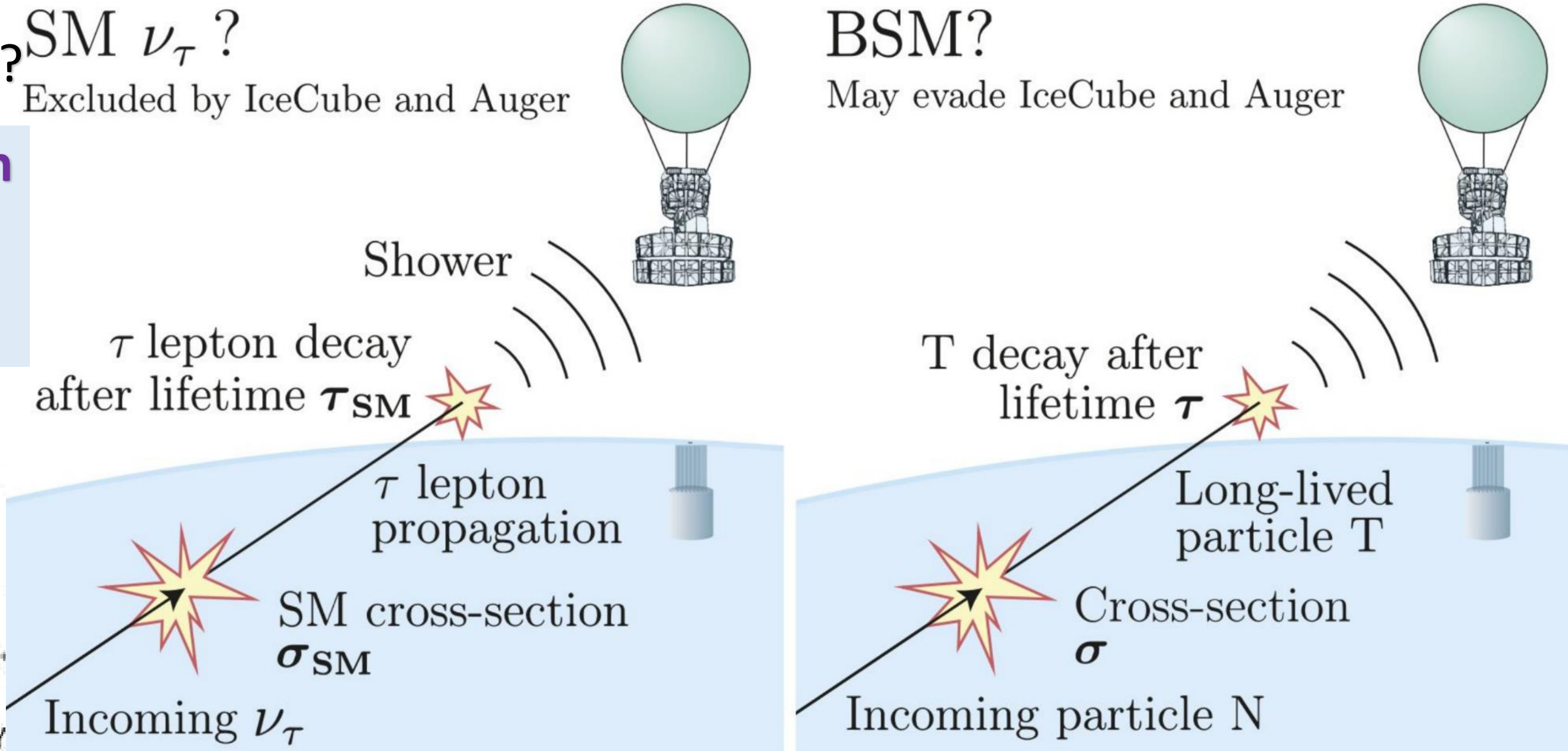
(IceCube, Auger)

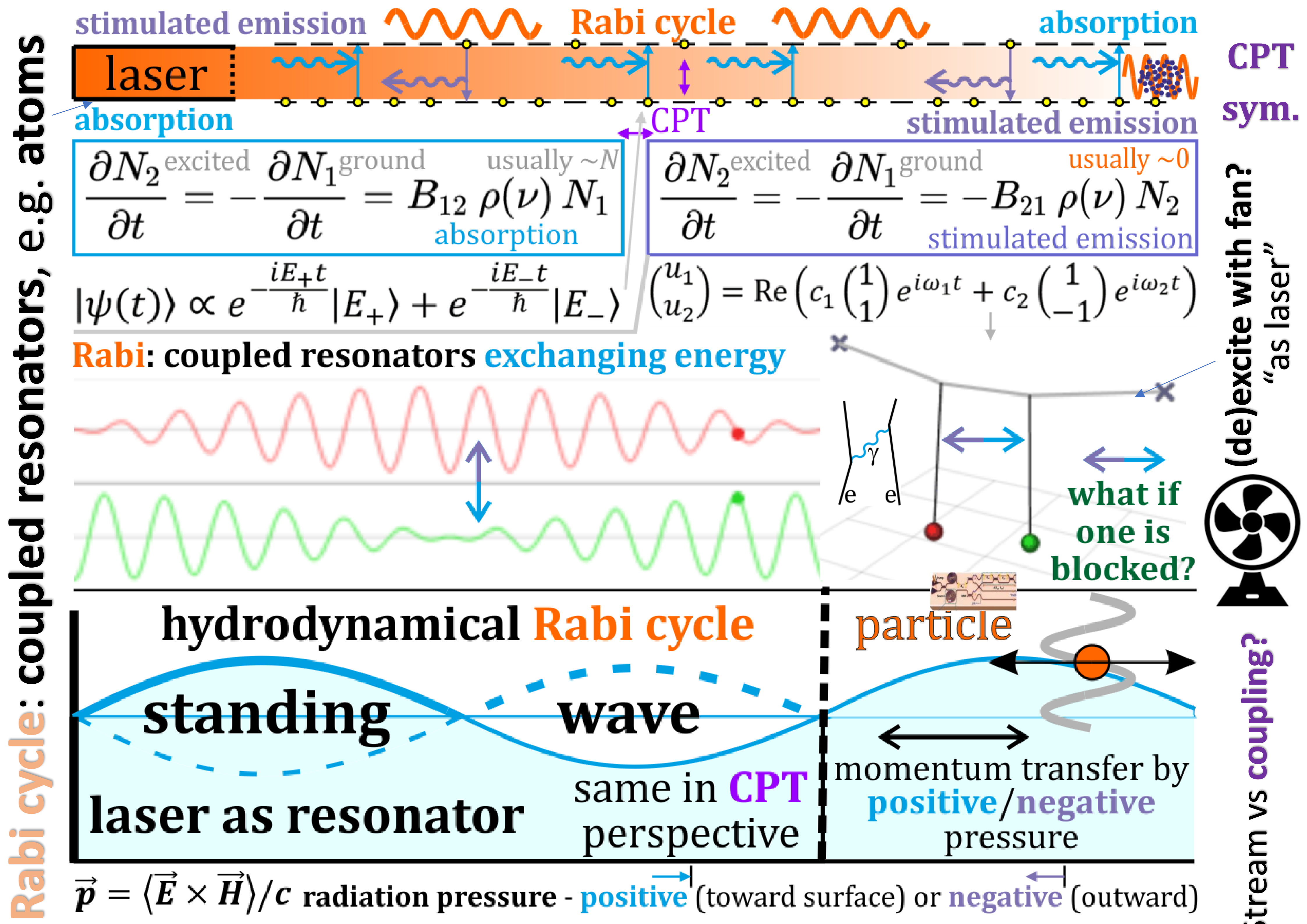
Needs long-lived after interaction???

Maybe some push scenario in CPT perspective?









Rabi cycle: coupled resonators, e.g. atoms
stimulated emission
Rabi cycle
absorption
laser
absorption
CPT
stimulated emission
CPT sym.
$\frac{\partial N_2^{\text{excited}}}{\partial t} = -\frac{\partial N_1^{\text{ground}}}{\partial t} = B_{12}\,\rho(\nu)\,N_1$
usually ~N
absorption
$\frac{\partial N_2^{\text{excited}}}{\partial t} = -\frac{\partial N_1^{\text{ground}}}{\partial t} = -B_{21}\,\rho(\nu)\,N_2$
usually ~0
stimulated emission
$|\psi(t)\rangle \propto e^{-\frac{iE_+ t}{\hbar}}|E_+\rangle + e^{-\frac{iE_- t}{\hbar}}|E_-\rangle$
$\begin{pmatrix}u_1\\u_2\end{pmatrix} = \mathrm{Re}\left(c_1\begin{pmatrix}1\\1\end{pmatrix}e^{i\omega_1 t} + c_2\begin{pmatrix}1\\-1\end{pmatrix}e^{i\omega_2 t}\right)$
Rabi: coupled resonators exchanging energy
γ
e
e
what if one is blocked?
(de)excite with fan?
"as laser"
hydrodynamical Rabi cycle
standing wave
laser as resonator
same in CPT perspective
particle
momentum transfer by positive/negative pressure
stream vs coupling?
$\vec{p} = \langle \vec{E} \times \vec{H} \rangle / c$ radiation pressure - positive (toward surface) or negative (outward)

Electrons in atoms... **Narrowing spectrum, coherence** **CPT perspective?**

Free electrons:

Synchrotron radiation:

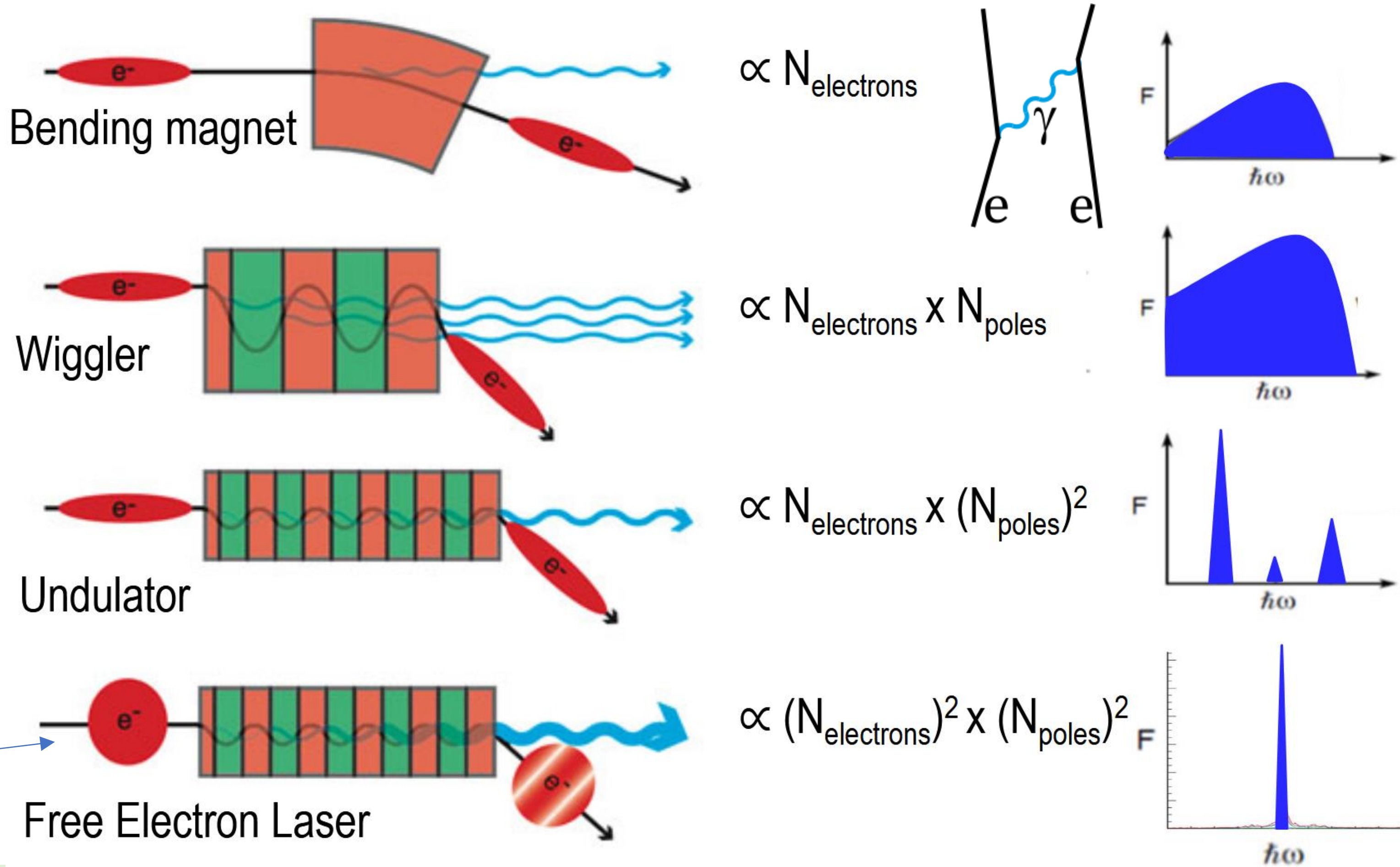
statistical **~spontaneous** emission: $P \propto N_e$

Coupled electrons ~superradiance:

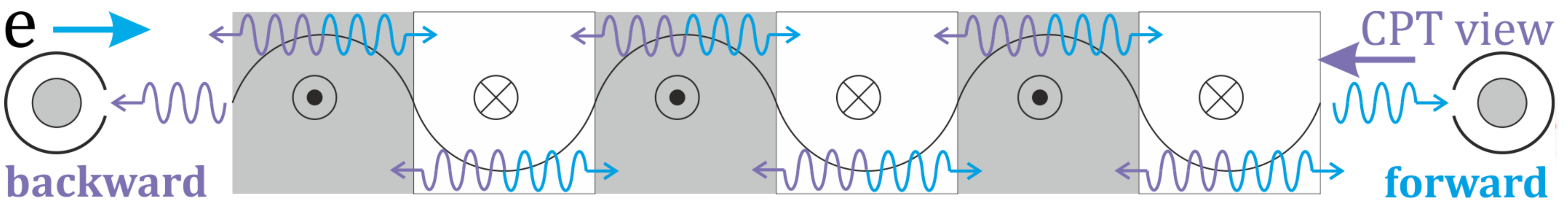
SASE in FEL, $P \propto N_e^2$

laser: **absorption↔ stimulated emission switched in T/CPT**

just **EM: T symmetric**



SASE: self-amplified spontaneous emission



wiggler/undulator/synchrotron: in CPT reversing electron & photon trajectory

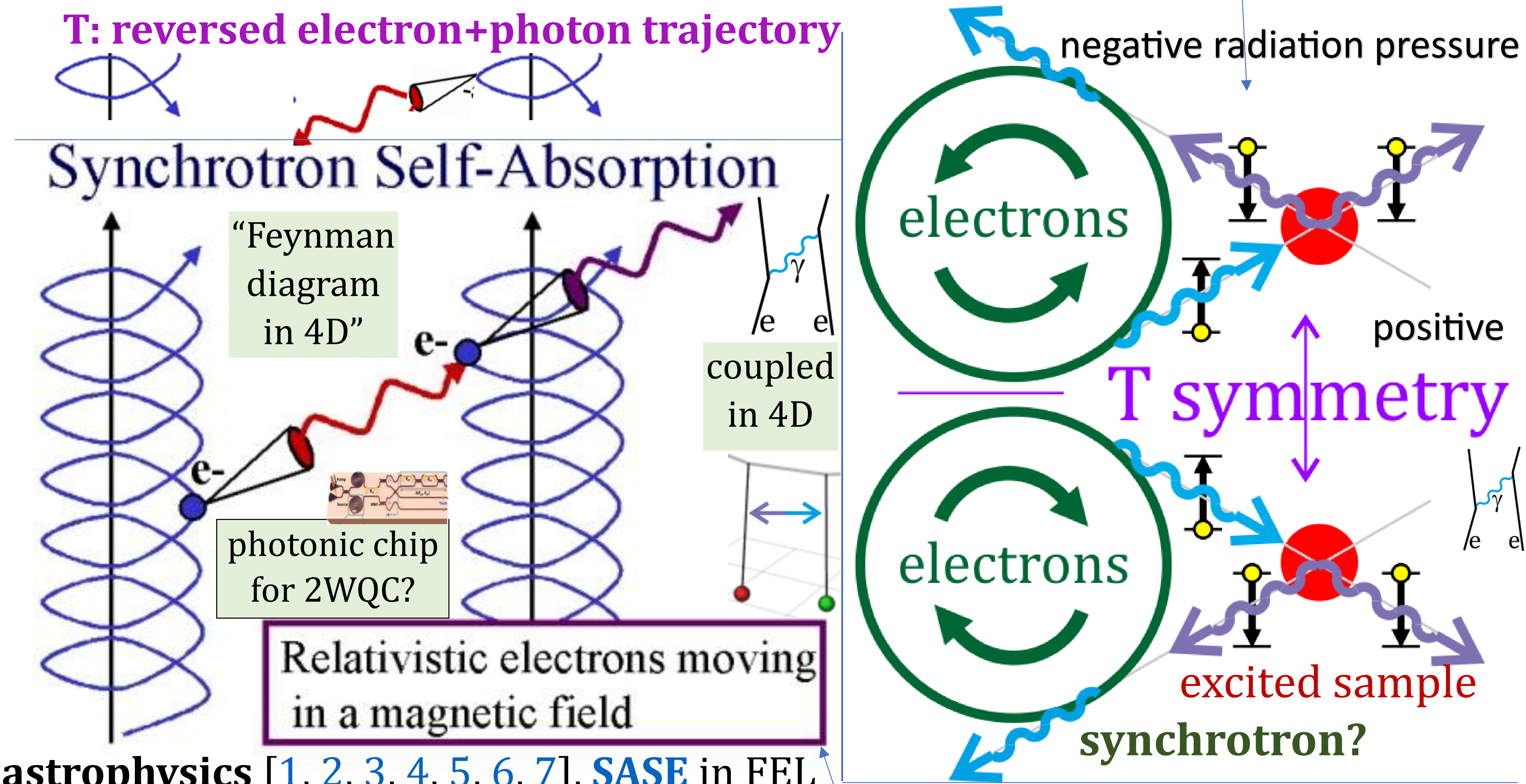
Synchrotron radiation by accelerating charge, also needed

CPT symmetric synchrotron self-absorption, self-amplified SE

What about **causality**? In **EM** should be **the same in T perspective**

Is there really spontaneous emission? Or coupled with e.g. electron?

T: reversed electron+photon trajectory

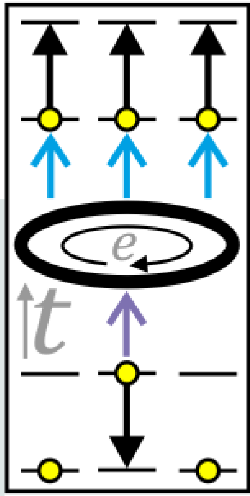


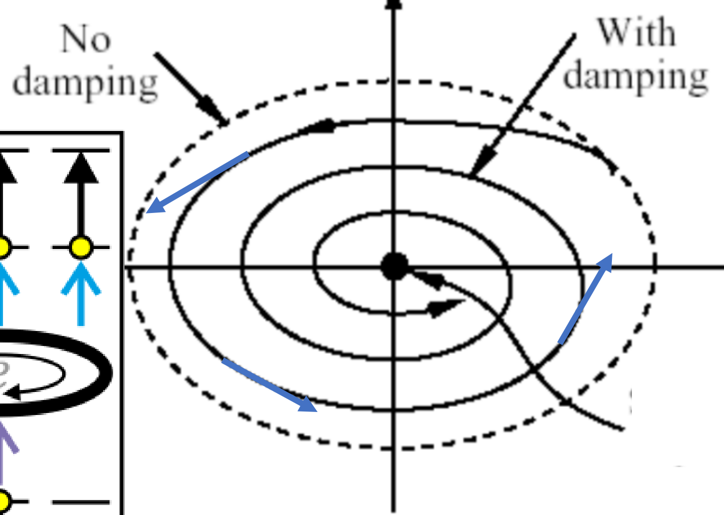
astrophysics [1, 2, 3, 4, 5, 6, 7], **SASE** in FEL

https://www.mssl.ucl.ac.uk/www_astro/lecturenotes/hea/radprocess/sld028.htm

Emission asymmetry CPT symmetry?

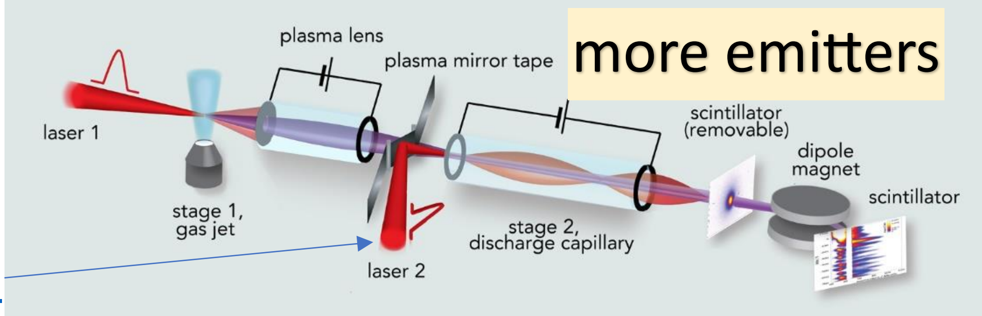
Circulating electron **loses energy** ([damping](#))
in **CPT** also circulating but **gains energy** instead.
Because of **more absorbers** than **emitters**?





Reversed:
gain energy
in [tabletop accelerators](#)

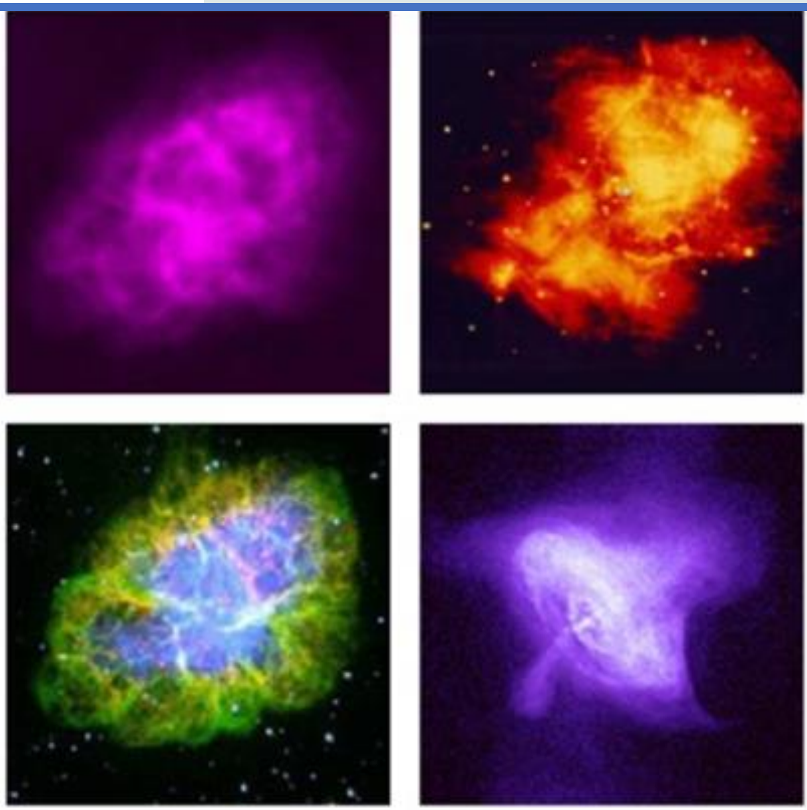
more emitters



In [CMB](#) spectrum?

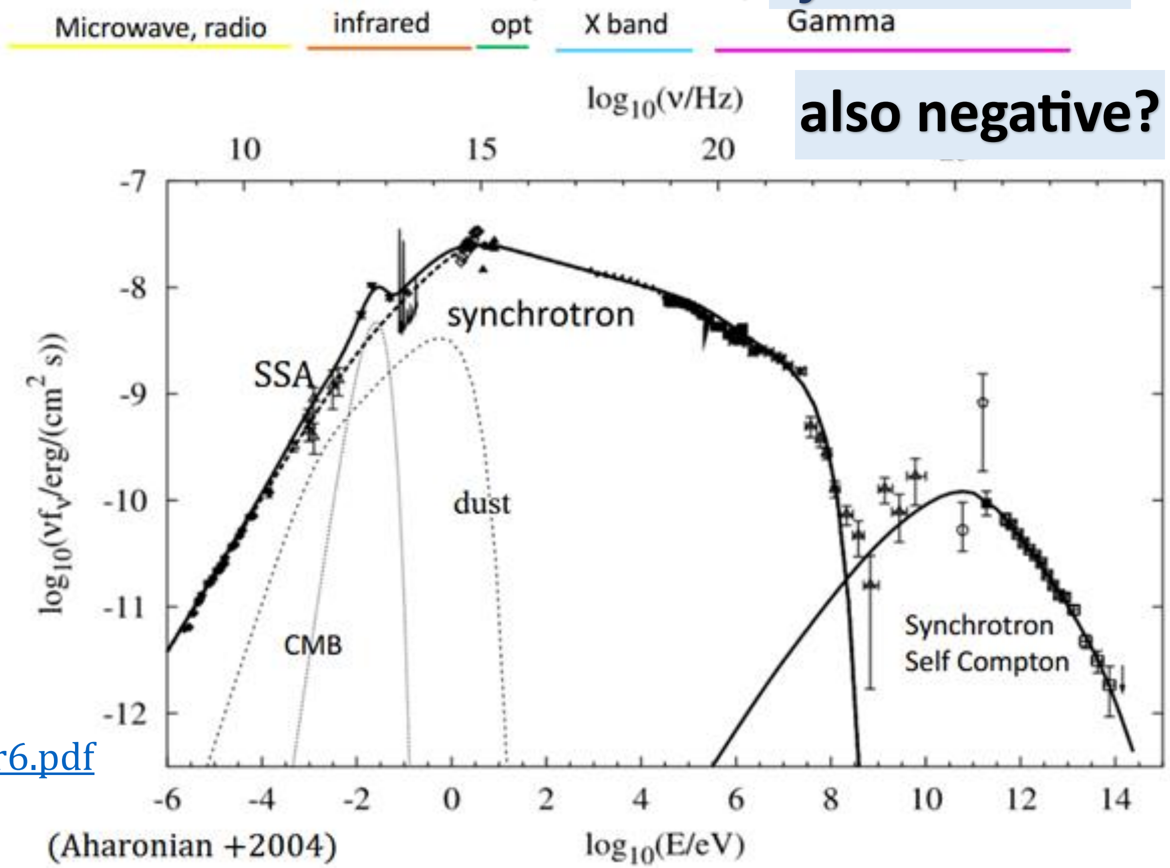
resonators? atoms, nuclei, synchrotron?

also negative?



https://people.sissa.it/~perrotta/lezioni_2023_2024/chapter6.pdf

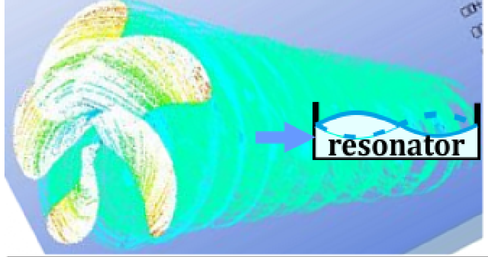
Images of the Crab in radio, infrared, optical and X-ray.
see http://chandra.harvard.edu/photo/0052/what.html.



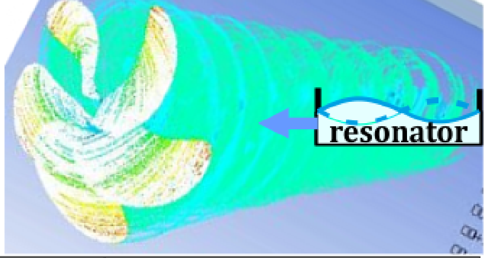
(Aharonian +2004)

Realizations?

Marine propeller-like antenna?



photon carries **energy**,
$\pm$ **momentum/pressure**,
$\pm$ **angular momentum**
like **swirl-like wave**
behind marine propeller



causing excitation
$\overset{T}{\leftrightarrow}$ **deexcitation**

T sym: reverse (radiation)

pressure: $\vec{p} = \langle \vec{E} \times \vec{H} \rangle / c$

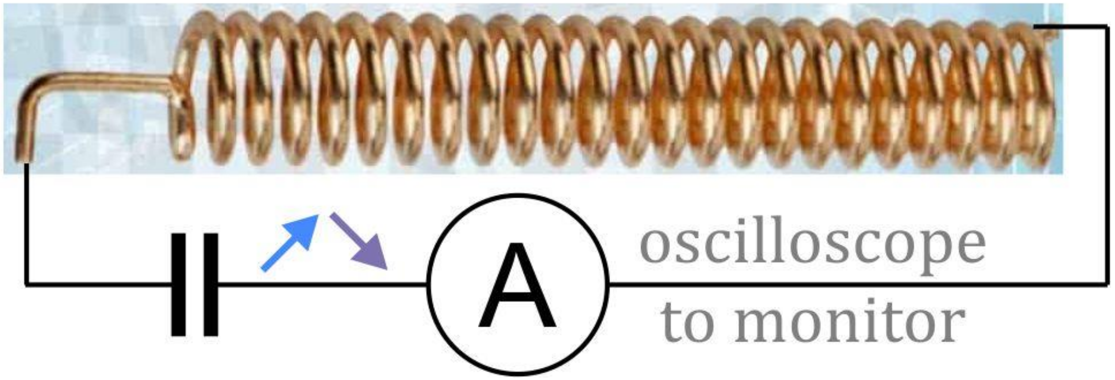
Theory	Gaug
Electro-dynamics	φ four-p
Hydro-dynamics	$\chi =$ flow v

First impulse to excite, like wireless
then **T-symmetric**: reversed voltage
impulse to get **faster relaxation**?

radio telescope with excited LC?

Reverse coupling?

transmitter: to **excite** with impulse → **receiver**: antenna + LC oscillator



T symmetry: reverse electron trajectory
also radiation pressure: $\vec{p} = \langle \vec{E} \times \vec{H} \rangle / c$
shouldn't **cause deexcitation** if excited?

ASE – amplified spontaneous emission (SASE: self-)

backward ASE **prevented by** **forward optical isolator**: **photons '→'**

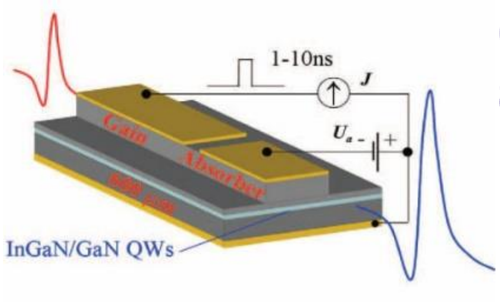
CPT? **forward ASE prevented by backward optical isolator? '←' ?**

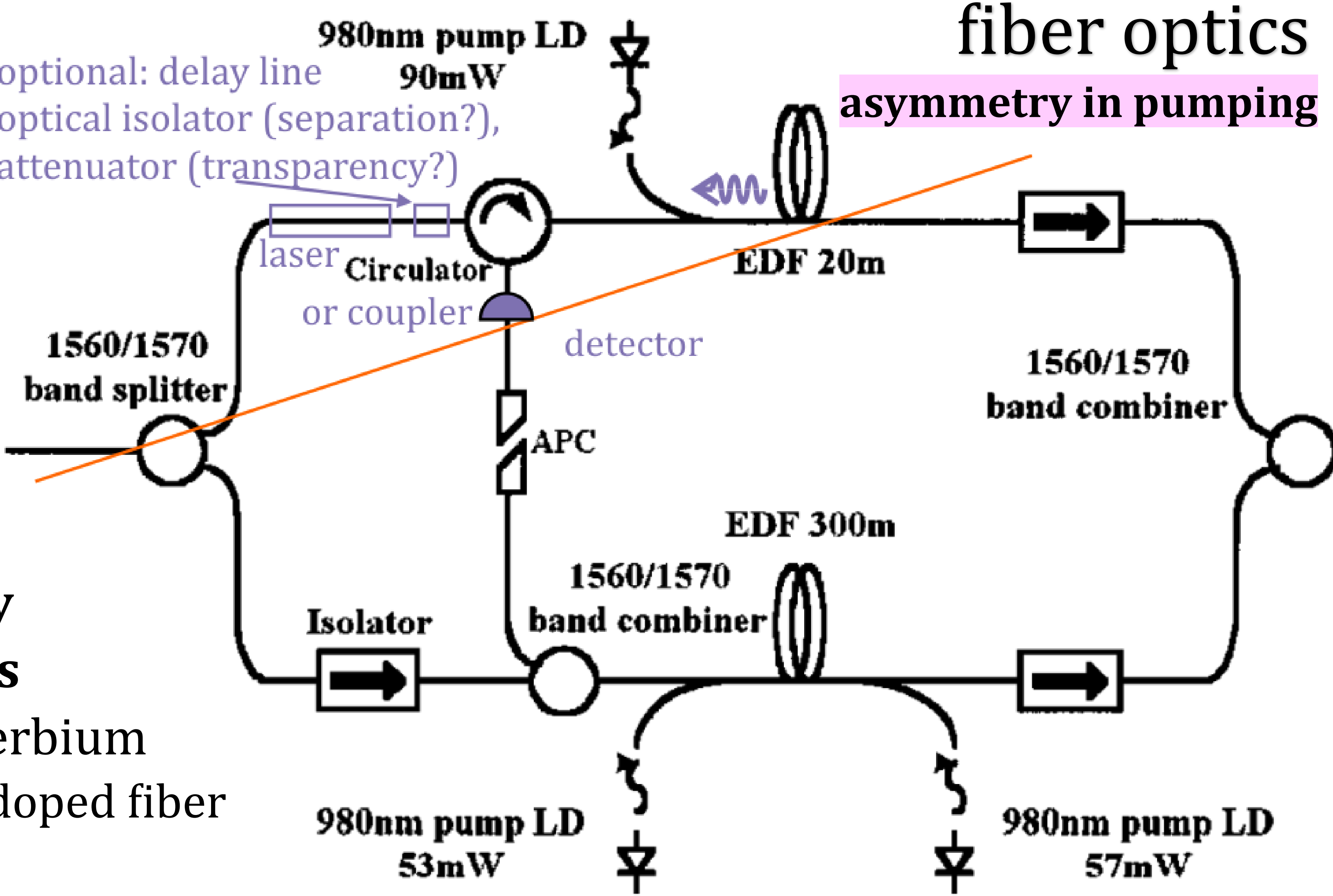
forward/backward ASE ~ positive/negative radiation pressure

fiber optics

asymmetry in pumping

optional: delay line
optical isolator (separation?),
attenuator (transparency?)



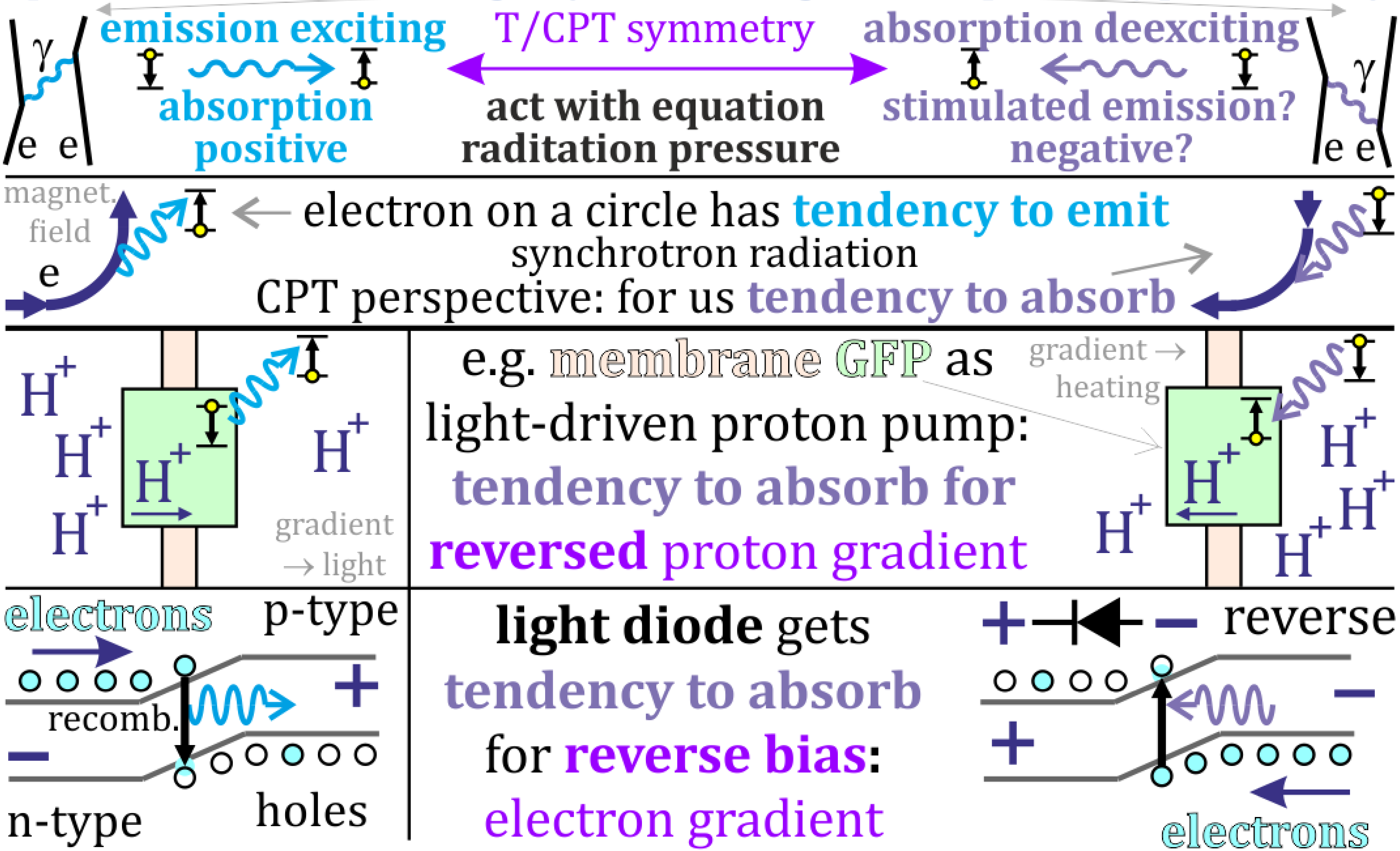


Black from
IEEE:841259
laser literally pulls photons
from excited erbium
EDF – erbium doped fiber

Various potential approaches for tendency to emit/absorb

S-matrix: $\langle\psi_f|U|\psi_i\rangle$ - e.g. emission/absorption **need both conditions**

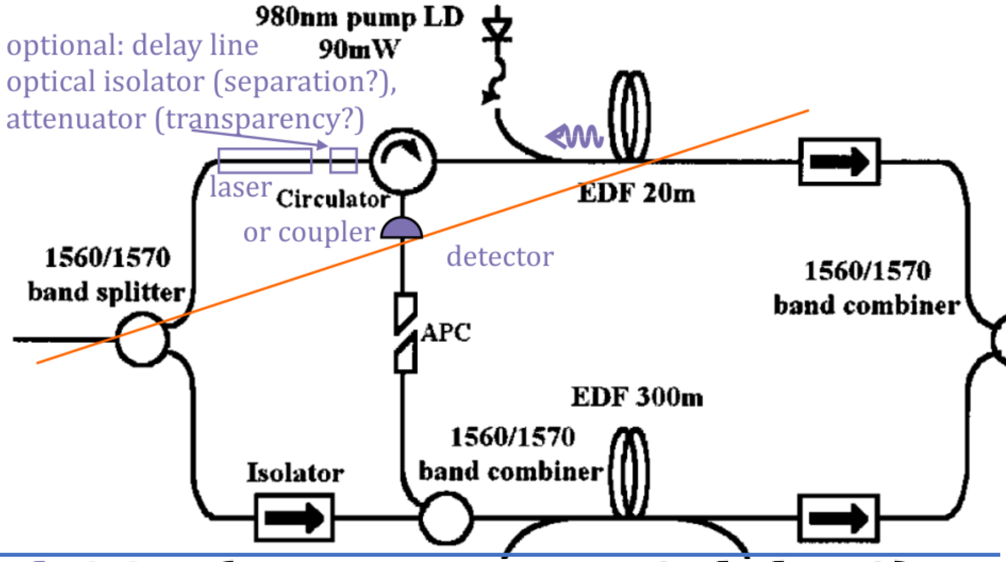
Spontaneous increasing Feynman diagram amplitude/probability



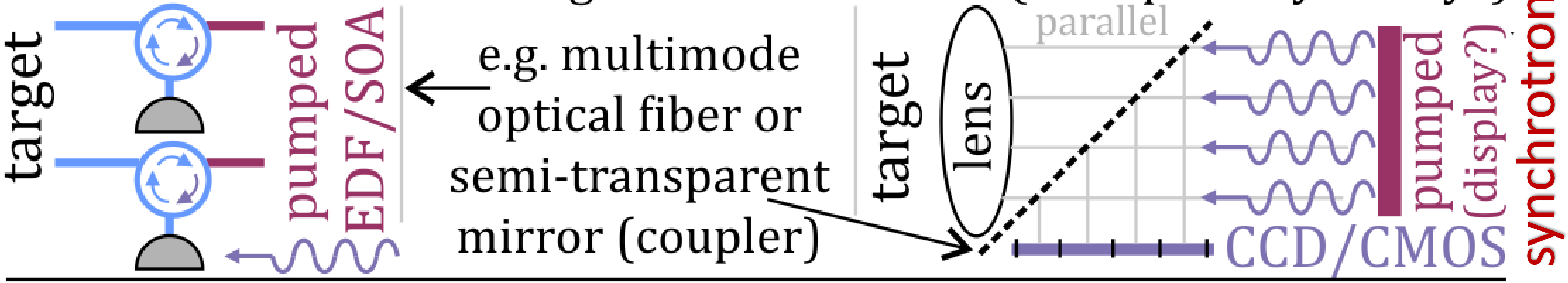
Detecting negative radiation pressure?

Backward camera? e.g. for emission CT, astronomy e.g. [pulsar synchrotron radiation]?

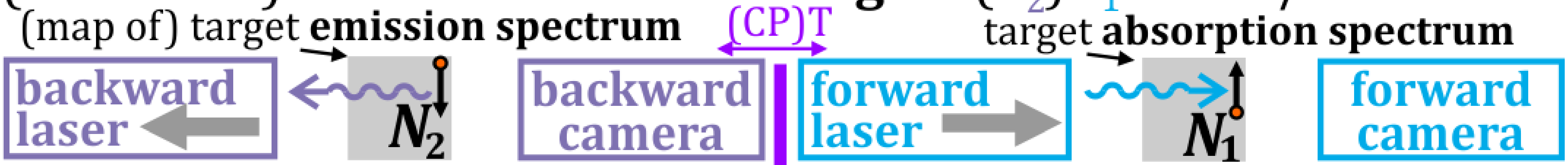
Backward lighted camera? amplify target deexcitation e.g. thermal, nuclear, attacks on e.g. [BB84] “pulling more photons”



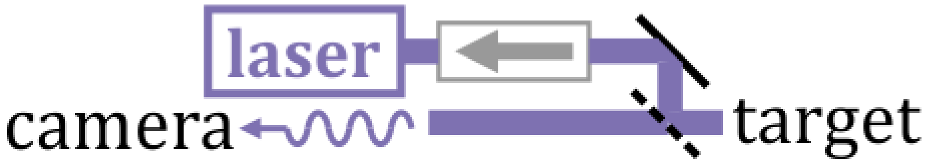
Backward camera: of target's backward ASE (transparency? delay?)



(backward) **transmission scanning**: of $(N_2) N_1$ atoms/molecules:



Backward lighted camera: causing backward ASE of target (BB84?)

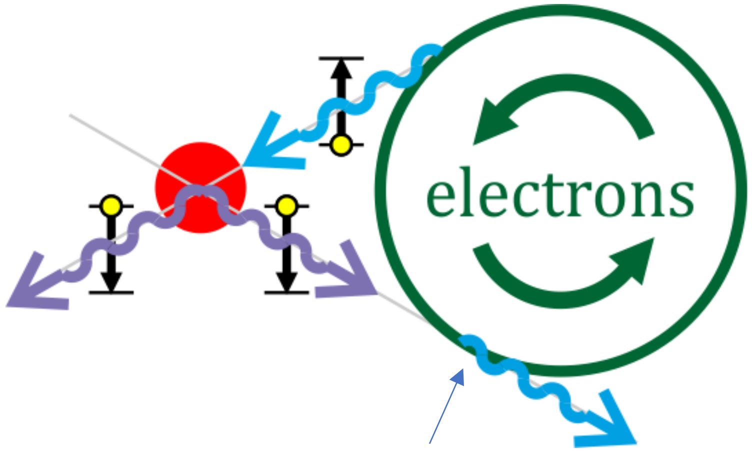


Amplifies target's directional emission e.g. of BB84 transmission laser (+CNOT)

Is absorption/ stimulated emission in agreement with CPT symmetry?

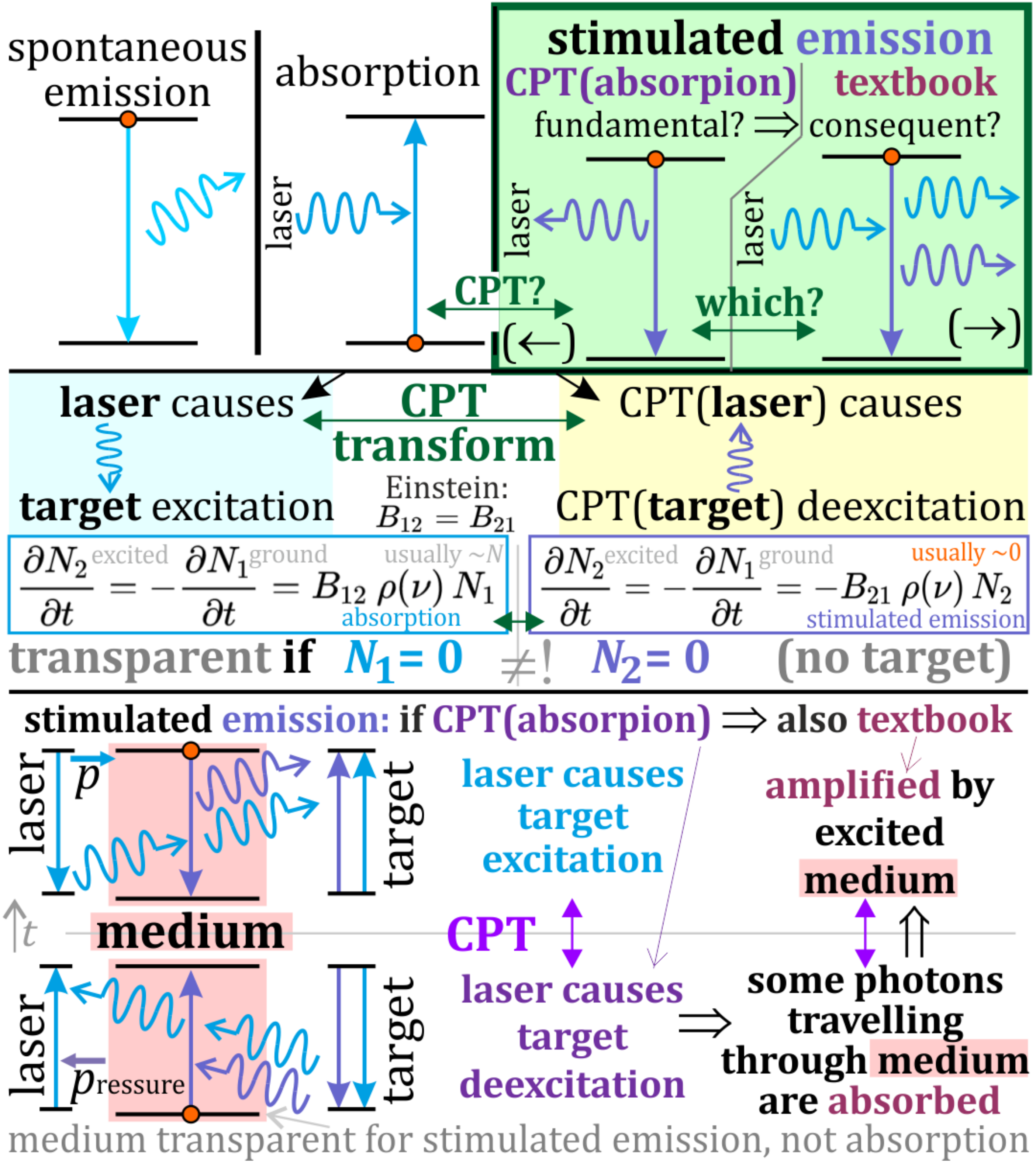
If not: **can we show macro CPT violation?**

Many **micro** CPT tests



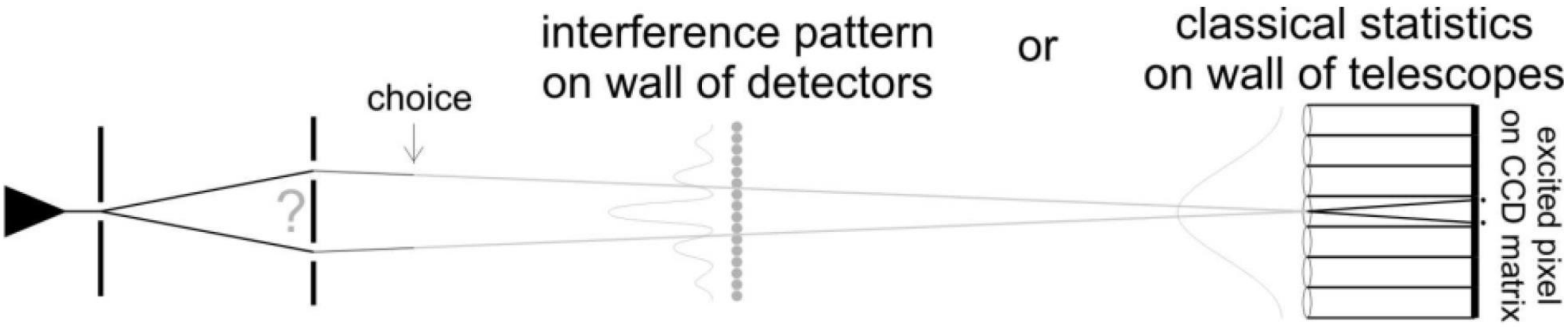
positive/negative radiation pressure: tendency to emit photons →/← (carried by photons?)

CT scanner modifications

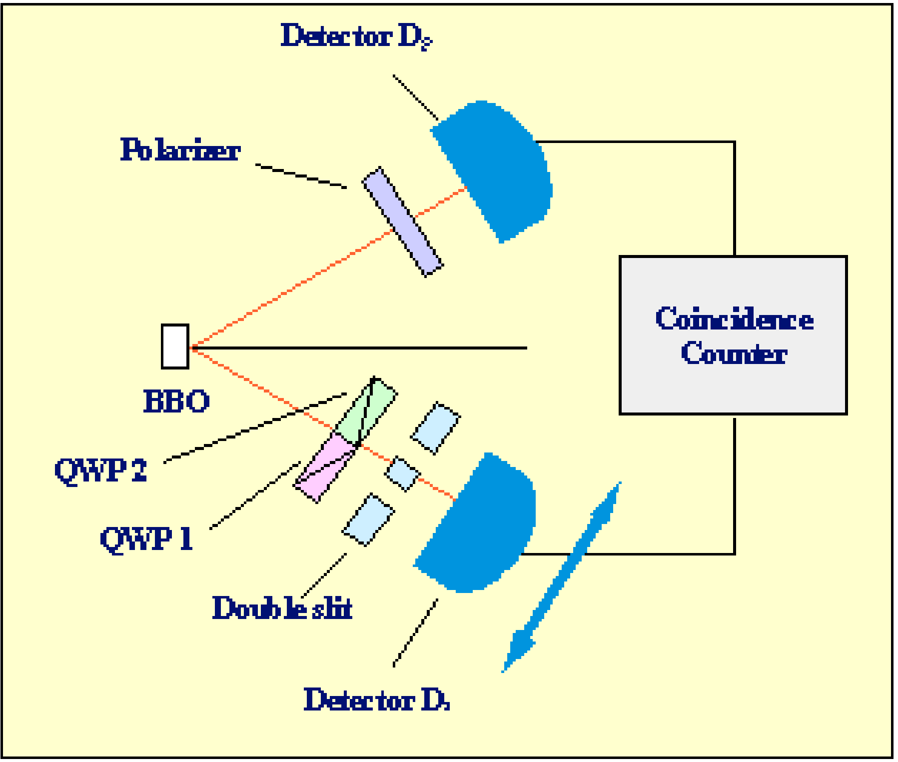


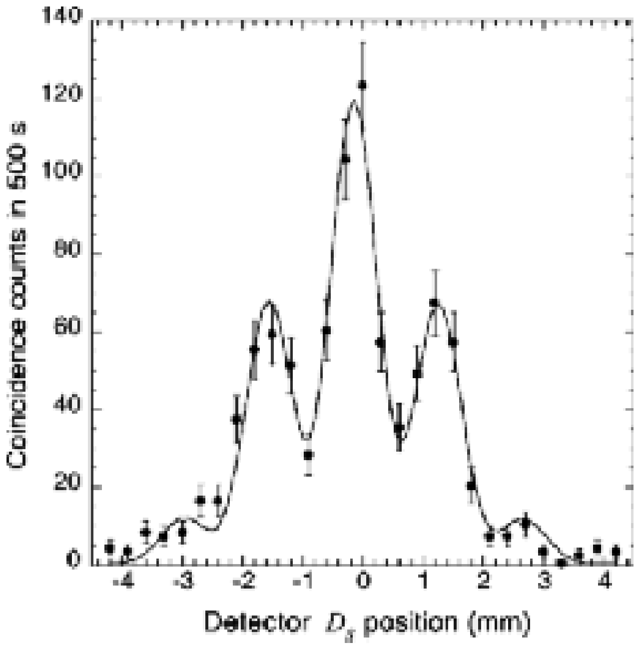
Time symmetry at heart of quantum … classical mechanics

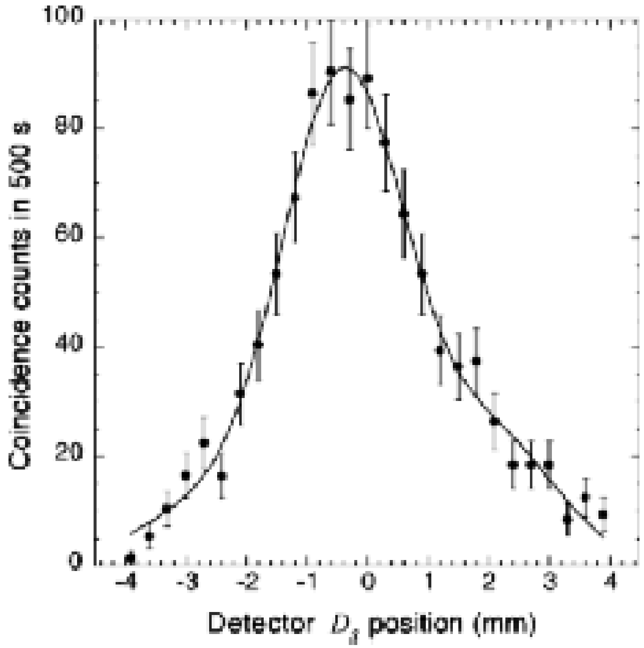
Wheeler experiment (Aspect's real.)– delayed choice classical/quantum



Double-slit quantum eraser (Walborn), explanation - **Rotating polarizer in one arm** we can erase which-path information, **changing second arm statistics**







Why **Born rule** in QM???
where its **square** comes from?

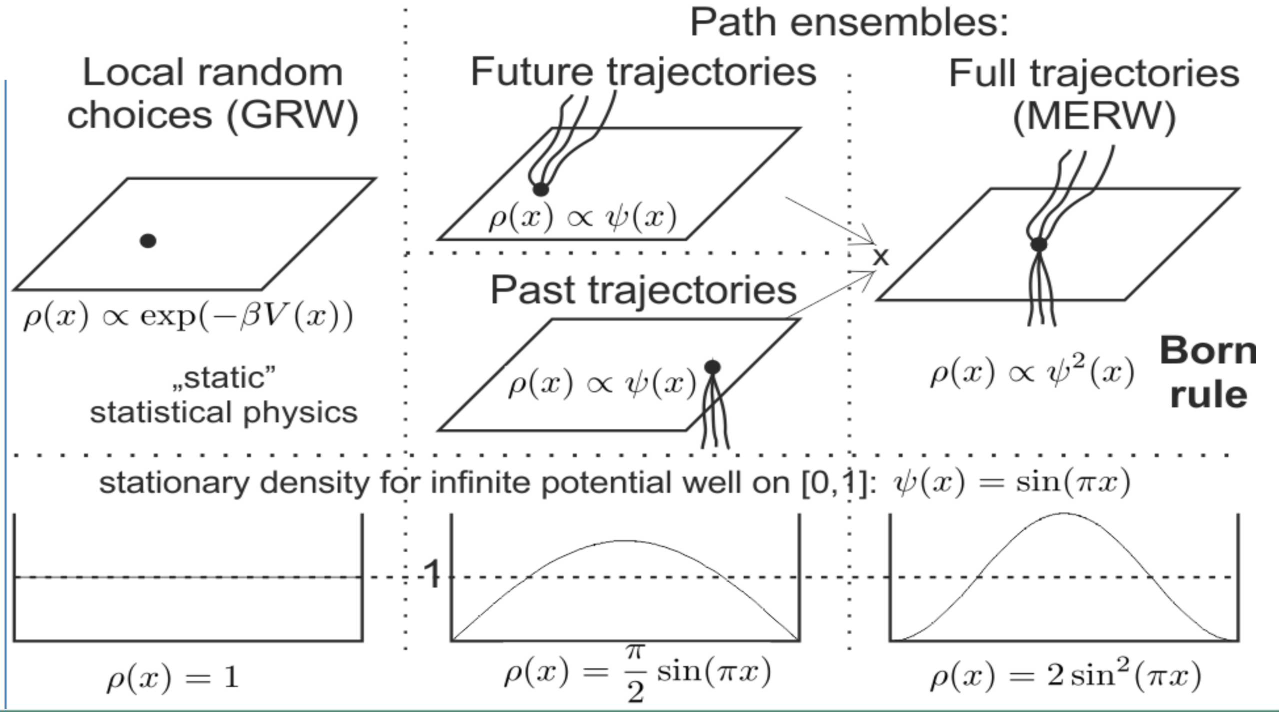
Scattering-matrix:

$$S_{\mathrm{fi}} = \lim_{t_2\to+\infty}\lim_{t_1\to-\infty}\langle\Phi_{\mathrm{f}}|\,U(t_2,t_1)\,|\Phi_{\mathrm{i}}\rangle$$

Two-state vector formalism:
propagators from
two time directions

Stationary distribution
in [0,1]? $\rho = 1$? $\rho \propto \sin^2$?
$\rho \propto \psi$ at the ends of Ising
(maximal entropy random walk)

Born allows violation of
Bell-like inequalities:
derived with Kolmogorov
3rd axiom instead



Mermin's inequality: „tossing 3 coins, at least 2 give the same"

$$\mathrm{P}(A=B) + \mathrm{P}(A=C) + \mathrm{P}(B=C) \boxed{\geq 1}$$

					+					+					=				
$A=0$	1		1			1	1				1			1		3	1	1	1
$A=1$		1		1				1	1		1			1		1	1	1	3
$B=$	0	1	0	1															
$C=$	0	0	1	1															

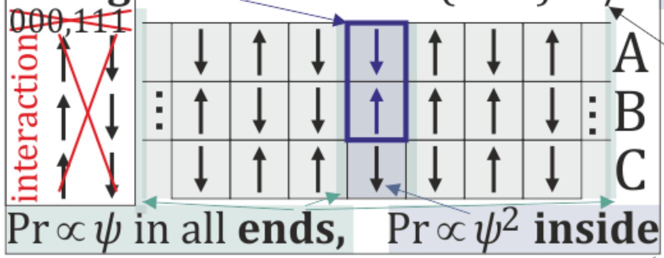
above: proof using **probabilities**

Kolmogorov 3rd axiom: $p_{AB?} = p_{AB0} + p_{AB1}$

Probability of disjoint events: sum of probabilitities vs $\propto$ squared sum of amplitudes

Born rule: $p_{AB?} \propto (\psi_{AB0} + \psi_{AB1})^2$

~Ising measure AB: Pr(A=B)=1/5



uniform path ensemble

MERW/Ising for **amplitudes**:

$$\psi = \begin{array}{|c|c|c|c|} \hline 0 & 1 & 1 & 1 \\ \hline 1 & 1 & 1 & 0 \\ \hline \end{array}$$

$\psi_{000} = \psi_{111} = 0$

violation with Born rule: $\mathrm{P}(A=B) = \dfrac{(\psi_{000}+\psi_{001})^2 + (\psi_{110}+\psi_{111})^2}{\sum_{A,B\in\{0,1\}}(\psi_{AB0}+\psi_{AB1})^2} = \dfrac{2}{10}$

$\mathrm{P}(A=B) + \mathrm{P}(A=C) + \mathrm{P}(B=C) \boxed{= 0.6}$

Inequality derived with 3rd axiom, might be violated with Born rule

both: impulse lasers

Observed negative delays

Science 2003, Neg. PRL 26

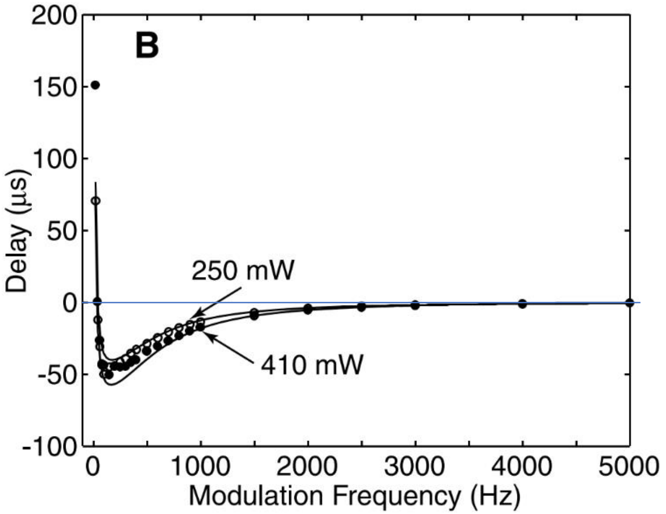
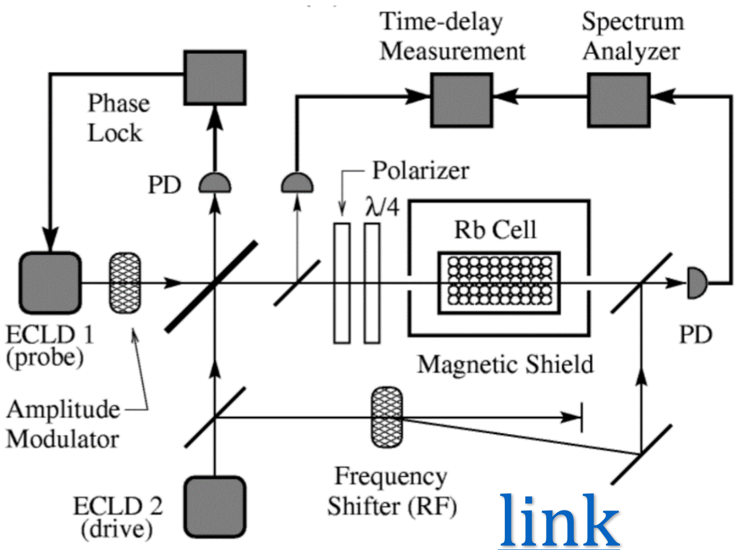


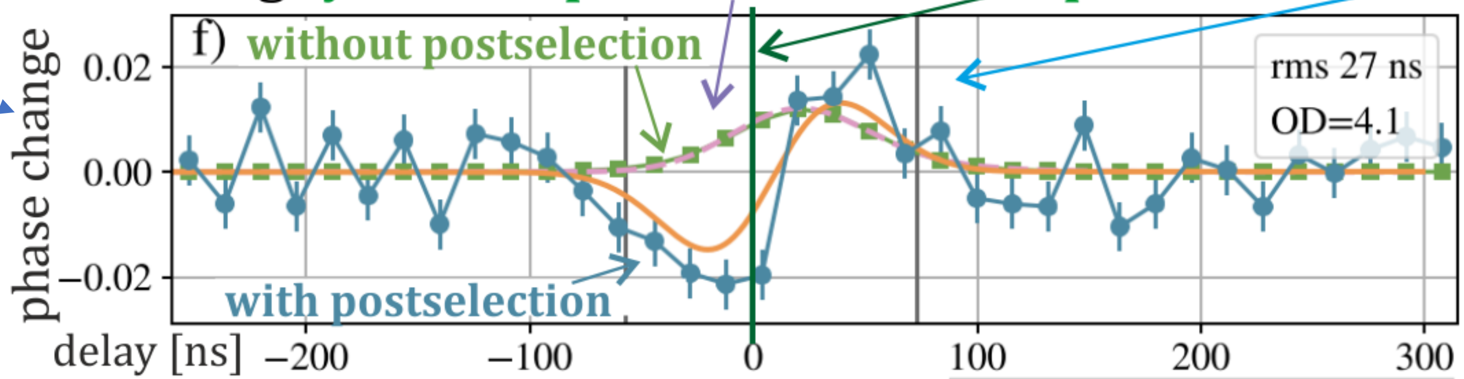
Fig. 2. (**A**) Relative modulation attenuation and (**B**) time delay measured for a 4-cm-long alexandrite crystal at a wavelength of 476 nm with pump powers of 250 and 410 mW. The observed negative time delay corresponds to superluminal propagation. The solid lines indicate the results of our theoretical model.

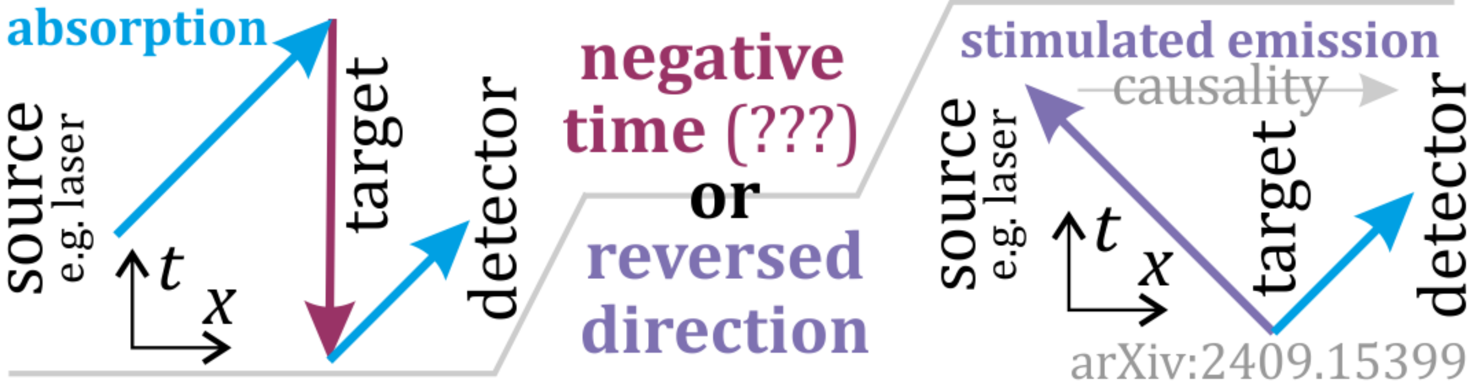


link

'Negative time evidence' Steinberg et al., arXiv:2409.03680

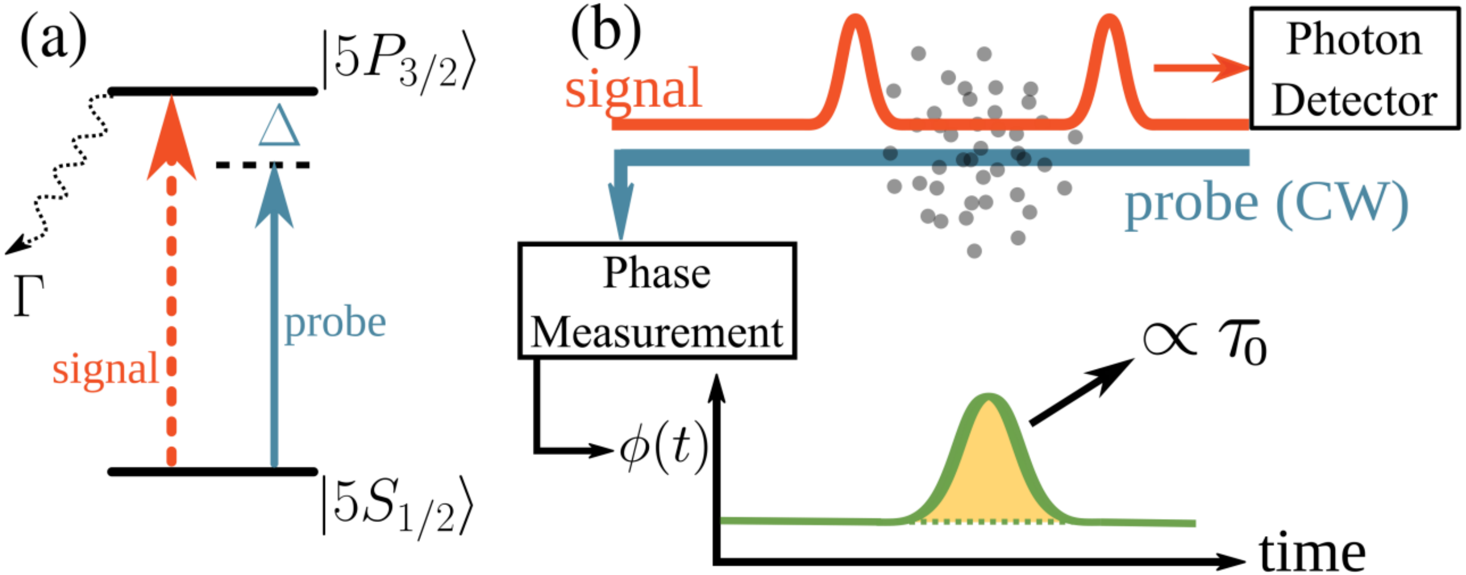
observing system response before the impulse and after





Proposed explanation: "**laser causes excitation/deexcitation**"

are **CPT symmetry analogs**: **should have opposite delay sign**



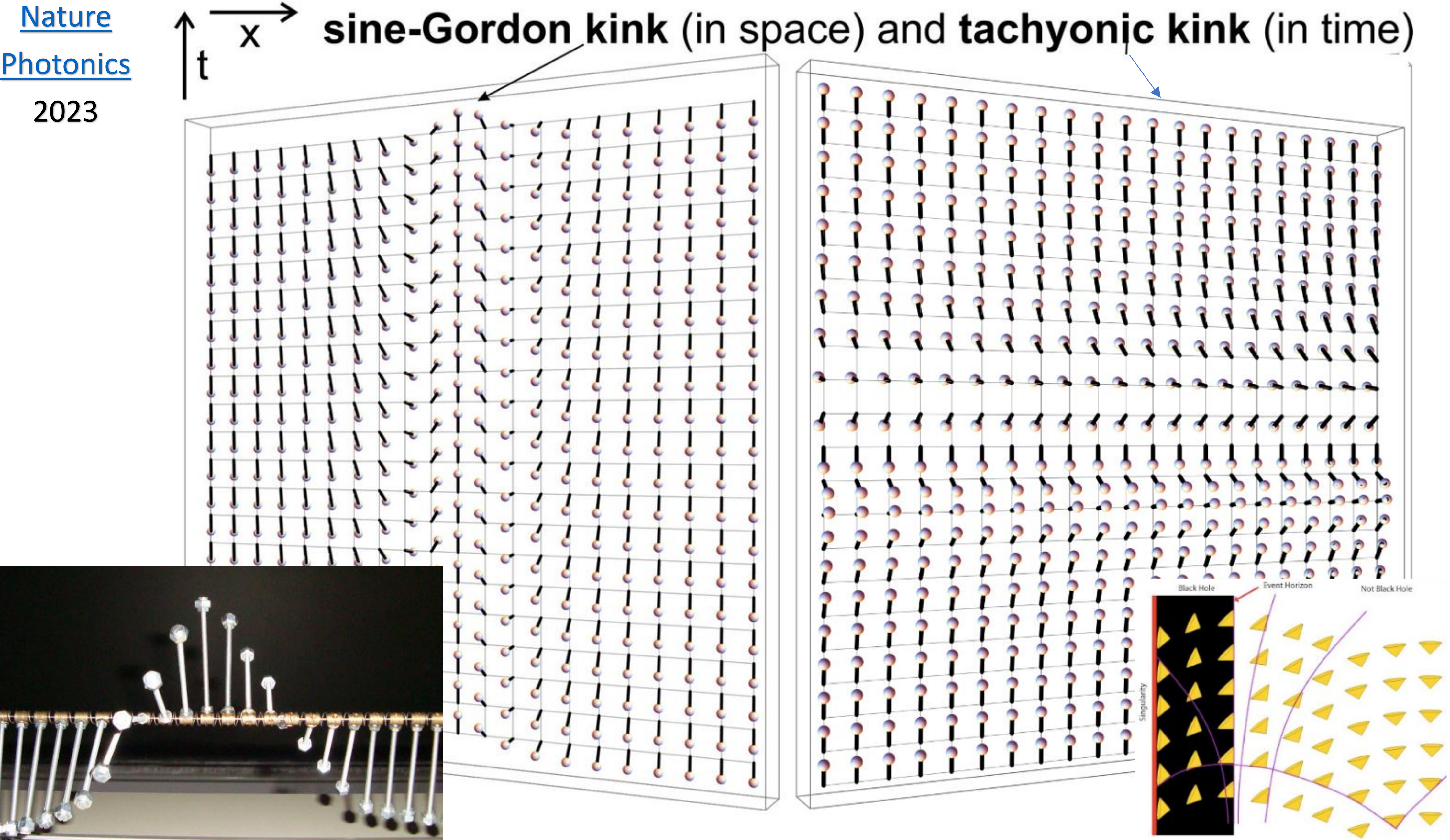
$\phi_{tt} - \phi_{xx} = \sin(\phi)$ kink solution in potential minimum $\xrightarrow[\text{BH?}]{x \leftrightarrow t}$ $\phi_{xx} - \phi_{tt} = -\sin(\phi)$ in maximum

Like **superradiance**: prepared excited medium, trigger: rapid burst quasiparticle … FTL?

FEL: "collective motion combined to create an electron Mexican wave that can **move faster than light**"

Nature Photonics 2023

sine-Gordon kink (in space) and **tachyonic kink** (in time)



(EPR) Rosen 1977: "Does gravitational radiation exist?"

retarded(-)/advanced(+) solutions, their **convex combinations**

Maxwell EM: $\square A_\mu = 4\pi J_\mu \quad \Rightarrow \quad A_\mu^\pm = \int \frac{J_\mu(\mathbf{r}', t \pm |\mathbf{r}-\mathbf{r}'|)}{|\mathbf{r}-\mathbf{r}'|} d^3\mathbf{r}'$

GW: $\square \tilde{h}_{\mu\nu} = -16\pi G T_{\mu\nu} \Rightarrow \tilde{h}_{\mu\nu}^\pm = 4G \int \frac{T_{\mu\nu}(\mathbf{r}', t \pm |\mathbf{r}-\mathbf{r}'|)}{|\mathbf{r}-\mathbf{r}'|} d^3\mathbf{r}'$

should depend on **boundary conditions**, like **emitters/absorbers**

Wheeler-Feynman assumed symmetric $\frac{1}{2}A_\mu^- + \frac{1}{2}A_\mu^+$, but instead

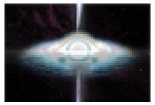
energy loss, inspiraling of orbiting show **Asymmetry of Radiation**

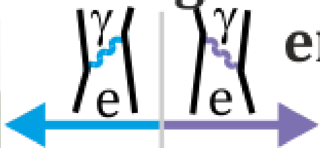
Current assumption: $\mathbf{1A_\mu^- + 0A_\mu^+}$ **only retarded** - verification???

why not e.g. $\mathbf{0.99A_\mu^- + 0.01A_\mu^+}$**?** GW can test (arXiv:2512.20692)

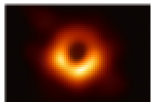
LIGO interferometer measures **length changes** - invariant to T/CPT symmetry, but **EM**:

absorption of **retarded** wave





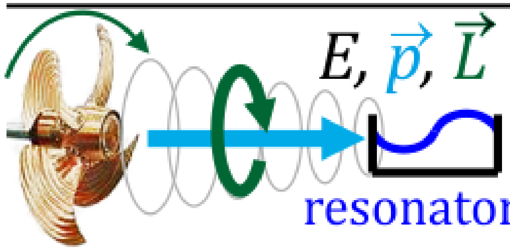
emission of **advanced** EM wave



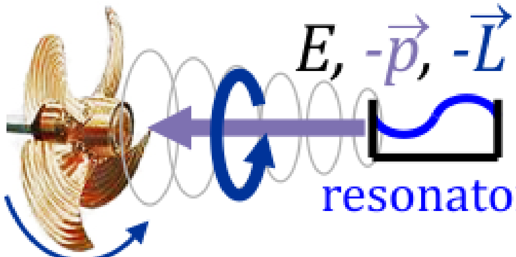
needs $N_1 > 0$ ground state

T symmetry

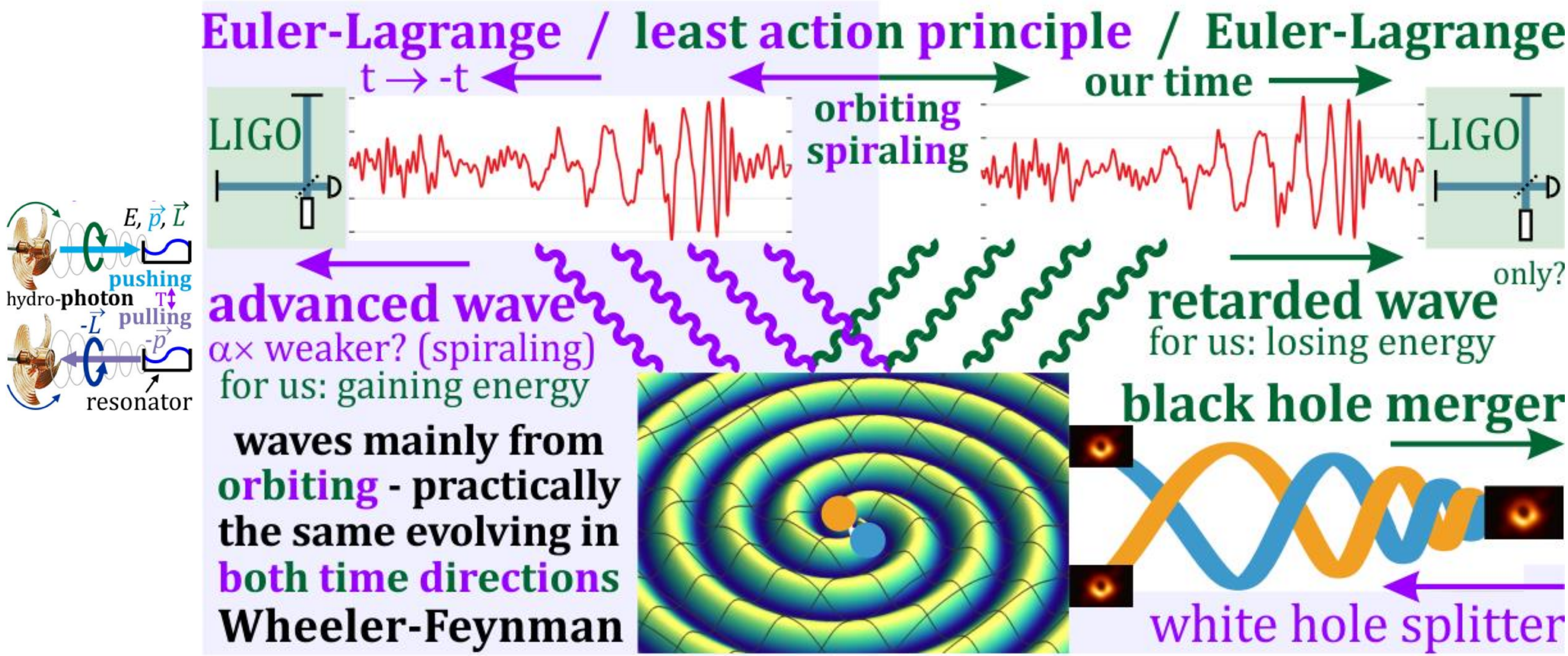
needs $N_2 > 0$ excited receiver



hydrodynamical analog: photon-like wave **pushing**/**pulling** energy from **resonator**, switched by T-symmetry: reversing rotation



LIGO interferometer measures **length changes** - invariant to T/CPT symmetry
should see **retarded** and **advanced waves** - **are there events for advanced?**



EM: $\square A_\mu = 4\pi J_\mu \quad \Rightarrow \quad A_\mu^\pm = \int \frac{J_\mu(\mathbf{r}', t \pm |\mathbf{r}-\mathbf{r}'|)}{|\mathbf{r}-\mathbf{r}'|} d^3\mathbf{r}'$ **retarded (-) / advanced (+)**

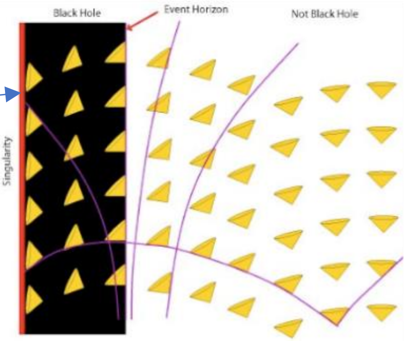
and

GW: $\square \tilde{h}_{\mu\nu} = -16\pi G T_{\mu\nu} \Rightarrow \tilde{h}_{\mu\nu}^\pm = 4G \int \frac{T_{\mu\nu}(\mathbf{r}', t \pm |\mathbf{r}-\mathbf{r}'|)}{|\mathbf{r}-\mathbf{r}'|} d^3\mathbf{r}'$ **convex combinations**

If **certain missing EM counterpart**, only **1** per ~**400** seen so far, consider **advanced wave**
Or "**too early events**" like 50-120 black hole Mass Gap for GW190521: **66** + **85** → **142** $M_\odot$?
Pulsar Timing Array see vibrations of the Universe: **require more** supermassive black holes
Observed **luminosity distance ~27 Gly** twice the **age of the Universe** - maybe advanced?

[arXiv: 2512. 20692]

https://en.wikipedia.org/wiki/**Non-orientable_wormhole**



General relativity in theory allows black hole horizon $t \leftrightarrow x$

Klein-bottle-like wormhole apply **T** symmetry to rocket

thought experiment

For external observer:

- **entropy decreases**, e.g. egg unscrambles,
- Reversed 1WQC,
- state $|0\rangle \leftrightarrow \langle 0|$,
- **pre-measurement**,
- **CT emission scan**,
- **retarded ↔advanced**
- **laser** causes deexcitation/**negative radiation pressure**,

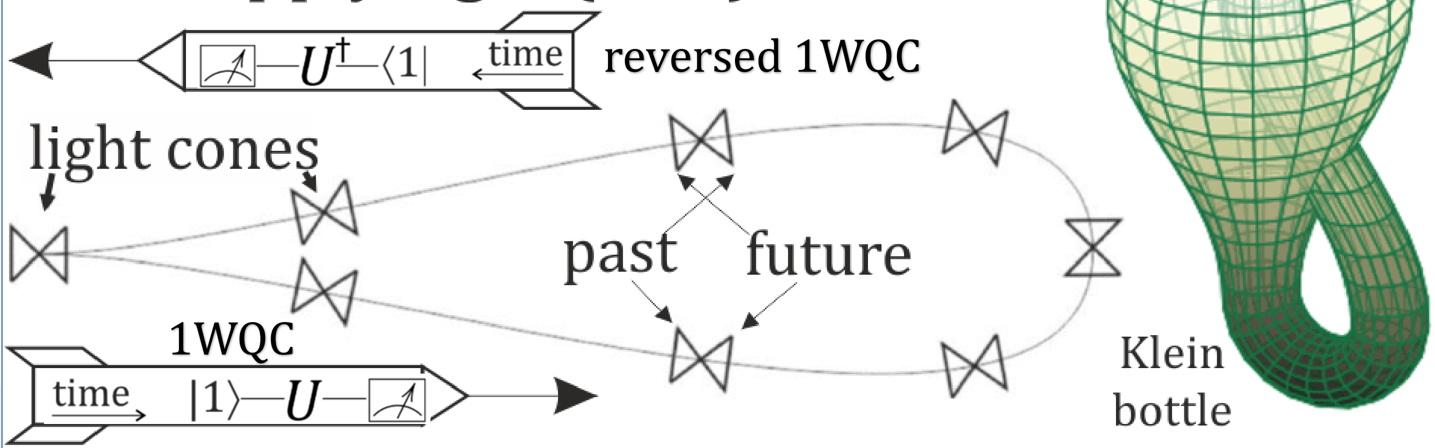
absorption↔emission

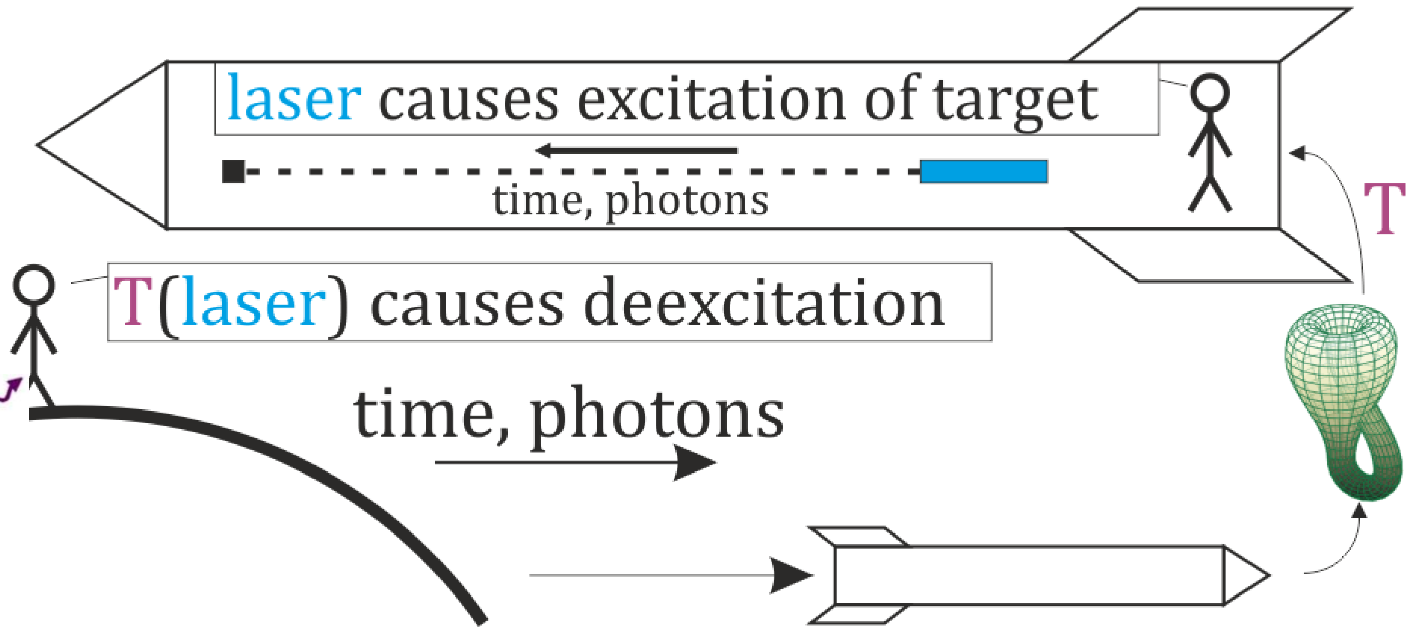
stimulated?

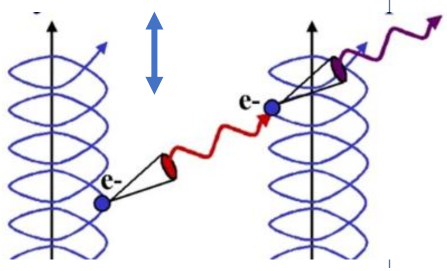
CPT
black ⟷ white hole

non-orientable wormhole

applying T (or P) transform

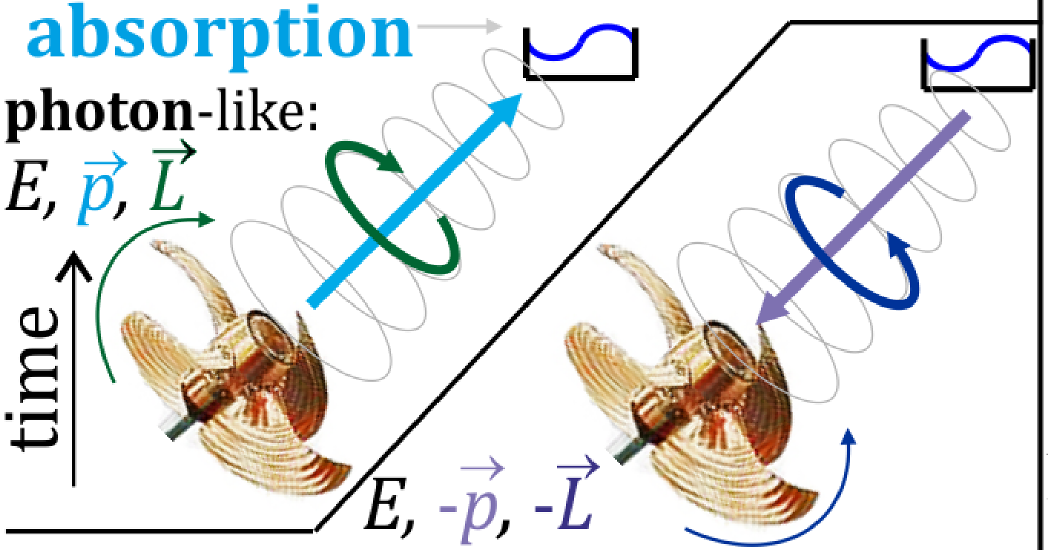






In theory 4 ways for hydrodynamics: what about EM?

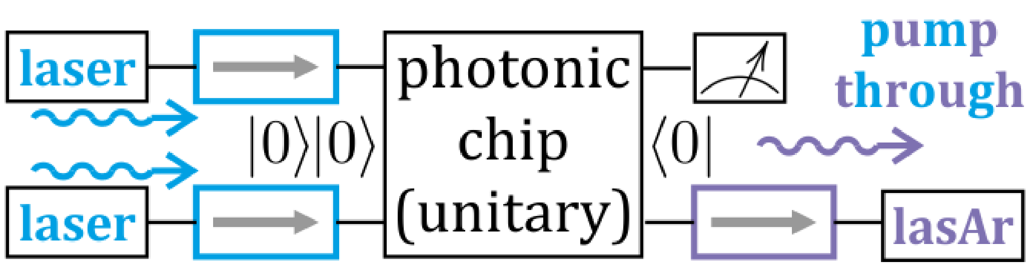
Retarded wave: positive delay
positive radiation pressure



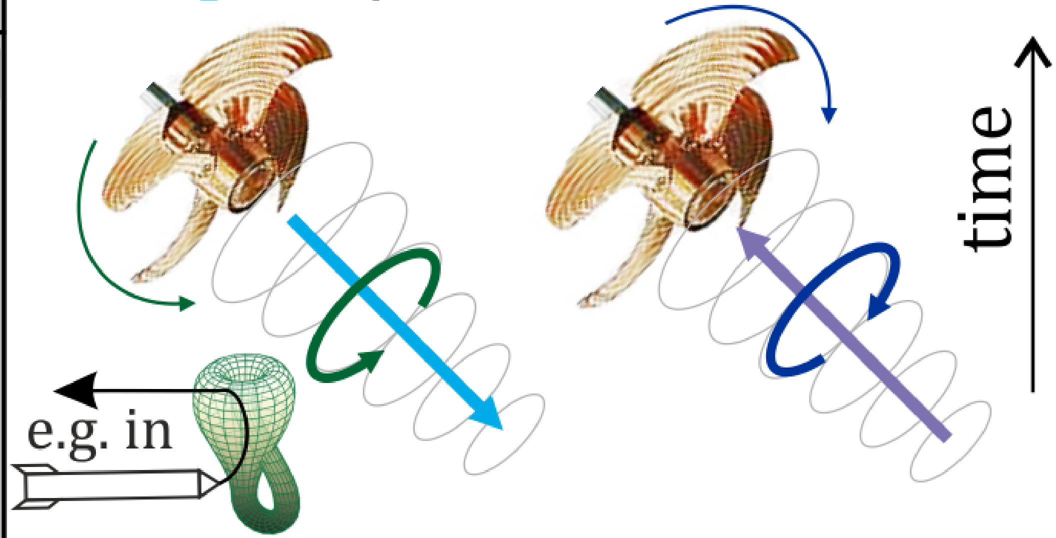
negative radiation pressure:
stimulated emission - 2WQC

solve postBQP∋ NP, needs quantum

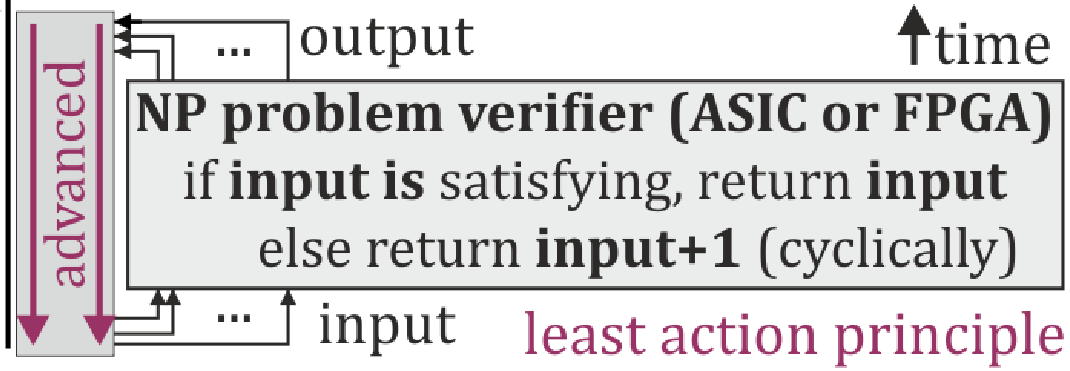
two-way control: **preparation** + **postparation**



Advanced: negative delay
absorption/stimulated emission



Time-loop computers: solving NP by augumenting classical computer, e.g. finding fixed point of a loop:



least action principle

Solve NP: negative rad. press. - quantum, advanced - classical

Let's speedup quantum supremacy! (e.g. solve NP, better error correction)

CPT symmetry: having $|0\rangle$, there is also $\langle 0|$

Jarek Duda, www.qaif.org/2wqc

2WQC: two-way quantum computers

adding $\langle 0|$ postparation: CPT(state preparation)

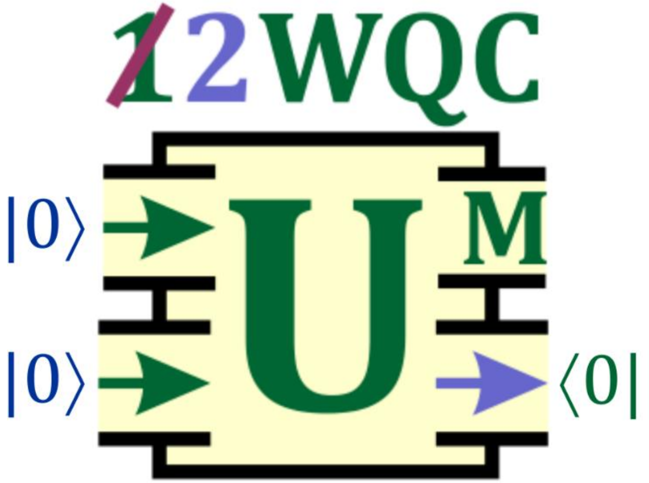
Acts as **postselection**, but with **higher success rate**

In CPT symmetry perspective use state preparation process

e.g. **low temperature**: $|0\rangle \leftrightarrow \langle 0|$

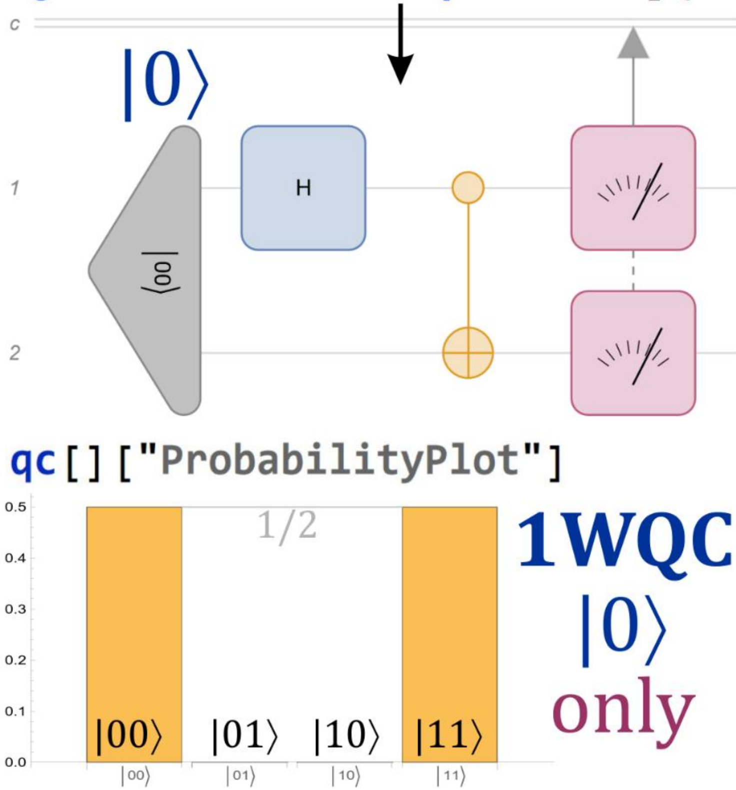
$$\langle\psi_f|U|\psi_i\rangle \overset{\text{CPT}}{\longleftrightarrow} \langle\psi_i|U^\dagger|\psi_f\rangle$$

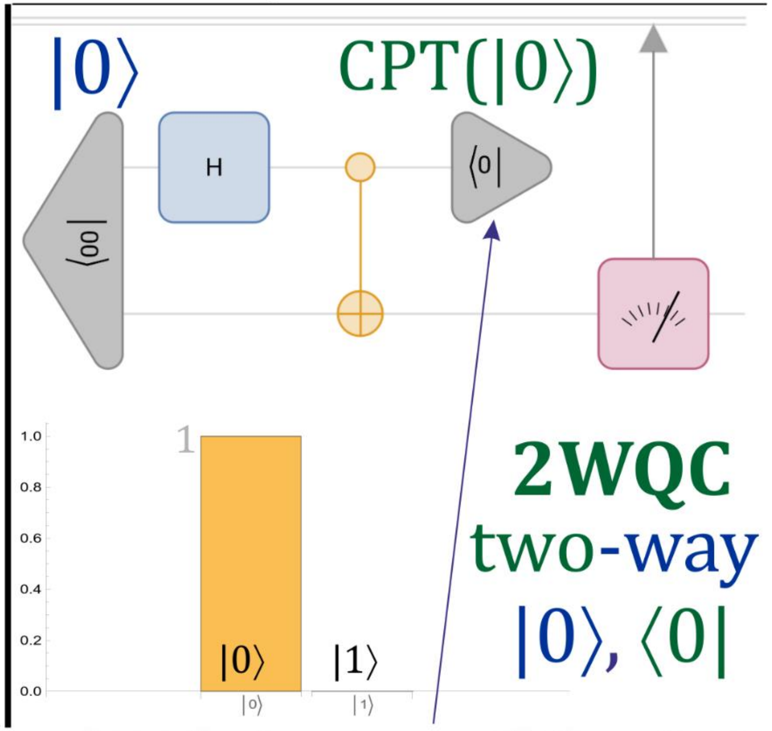
Evolve forward $\leftrightarrow$ backward




```
QuantumCircuitOperator[{"00", "H" → 1, "CNOT", {1, 2}}]
```

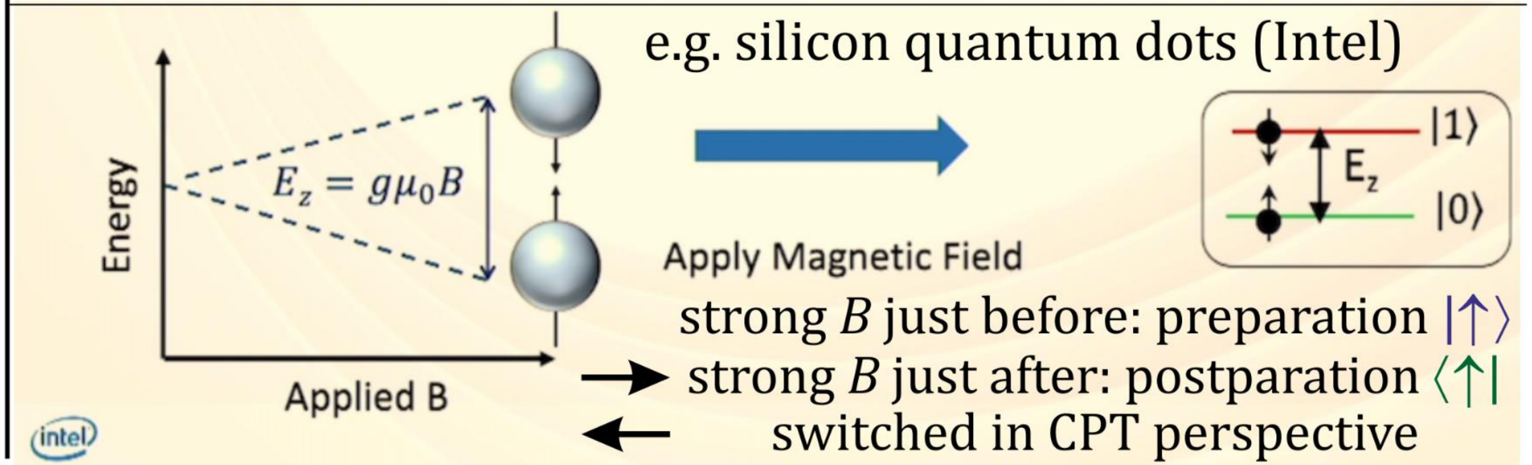
Wolfram Quantum Framework






```
QuantumCircuitOperator[{"00", "H" → 1, "CNOT", SuperDagger["0"], {2}}]
```

e.g. silicon quantum dots (Intel)



strong B just before: preparation $|\uparrow\rangle$

strong B just after: postparation $\langle\uparrow|$

switched in CPT perspective

Pre: [https://en.wikipedia.org/wiki/**Maximal_entropy_random_walk**](https://en.wikipedia.org/wiki/Maximal_entropy_random_walk), *arXiv:0910.2724*

2023: *arXiv:2308.13522* "Two-way quantum computers adding CPT analog of state preparation"

2024: **2WQC XPRIZE team**, **~40 QInterns** … **NP solver, better error correction**

G. Czelusta, **Grover's algorithm** on two-way quantum computer, *arXiv:2406.09450*

M. Noor, J. Duda, **No-cloning theorem** for 2WQC and postselection, *arXiv:2407.15623*

J. Duda, **3-SAT solver** for two-way quantum computers, *arXiv:2408.05812*

A. Linden, B. Gül, Optimization of **postselection** … 2WQC approach, *arXiv:2409.03785*

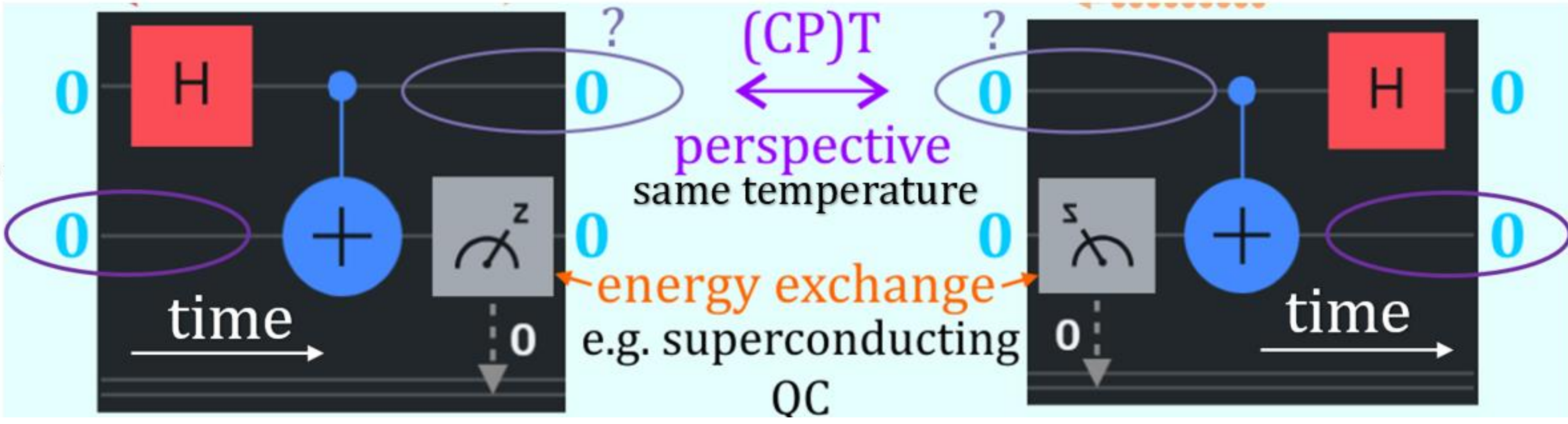
Current talk: J. Duda, Testing stimulated emission photon direction, *arXiv:2409.15399*

Many arguments, no counter? **search for help with experimental confirmation**

e.g. |0⟩ by thermalization: nature.com/articles/s41598-025-87323-x

"the system is dissipative and decohering in both temporal directions"

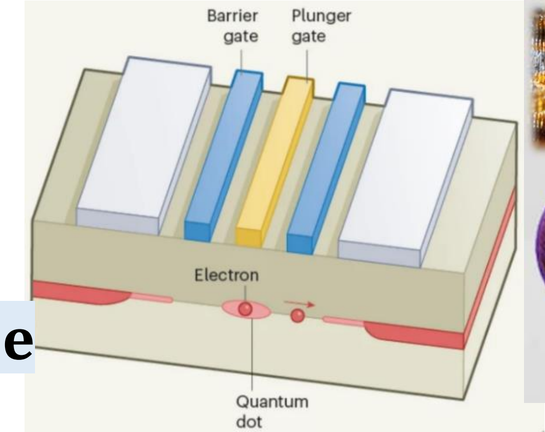
~30mK low temperature reseirvoir for ~|0⟩⟨0| state preparation (thermalize)

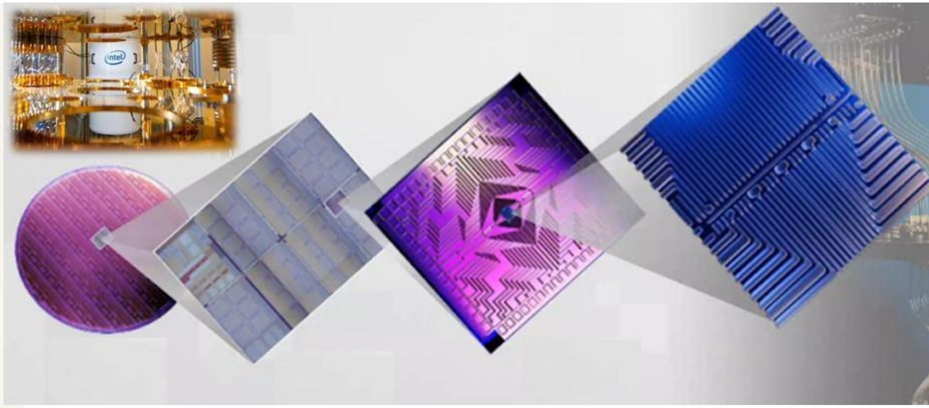


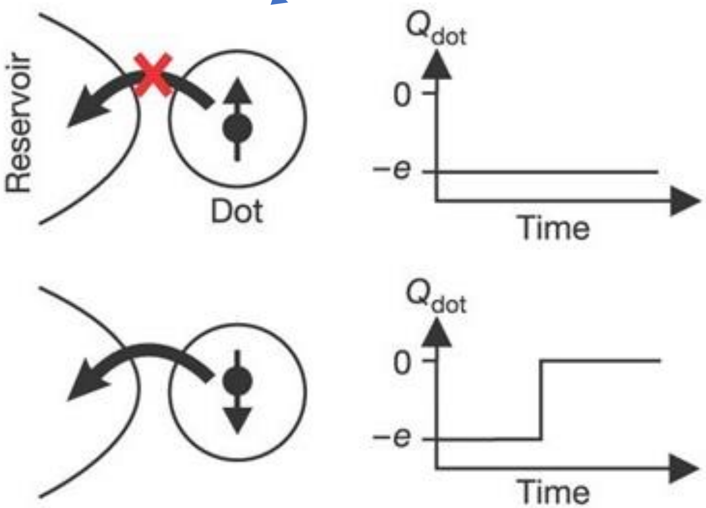
Silicon quantum dots

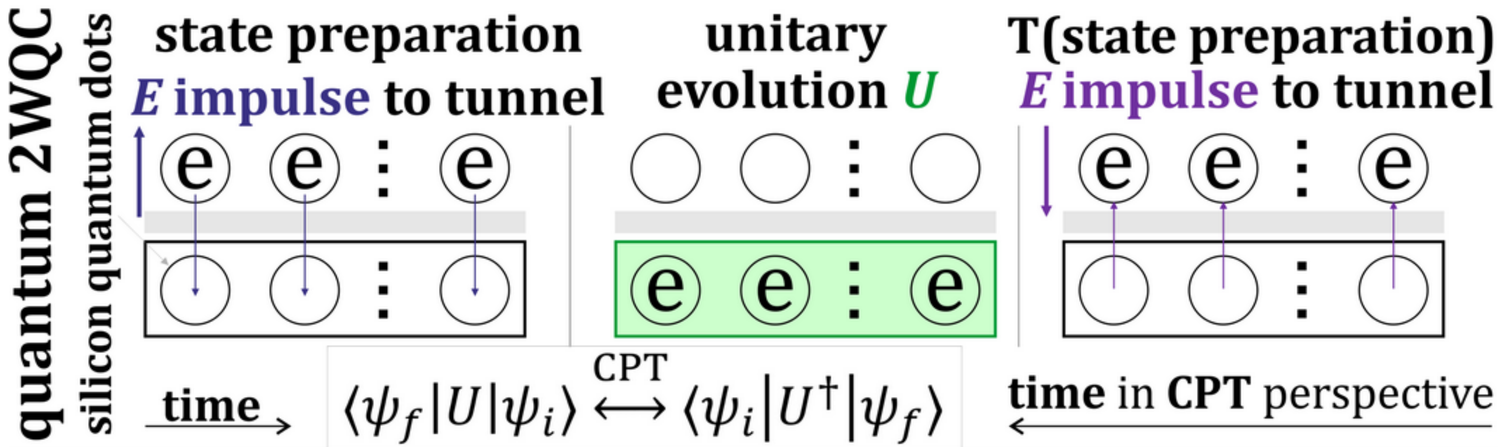
e.g. Intel 12 qubit

All operations with
EM fields – easy to reverse
spin or position qubits





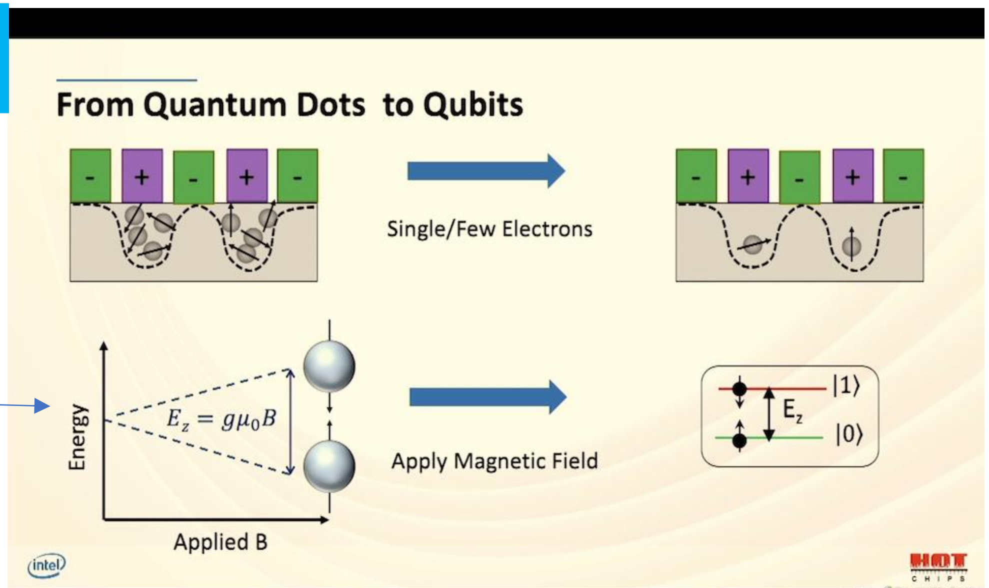




reverse applied impulse: $V(t) \leftrightarrow V(-t)$

Elzerman readout: only ↓ spin can tunnel
in Intel 2024 article for state preparation

Or use magnetic field to enforce spin direction
before for $|0\rangle$, opposite after for $\langle 0|$



better stability thanks to 2WQC two-way flow control

arXiv:2406.09450 **Grover's algorithm** on two-way quantum computer

(G. Czelusta) in $O(1)$ time instead of $O(\sqrt{n})$, **more error resistant**

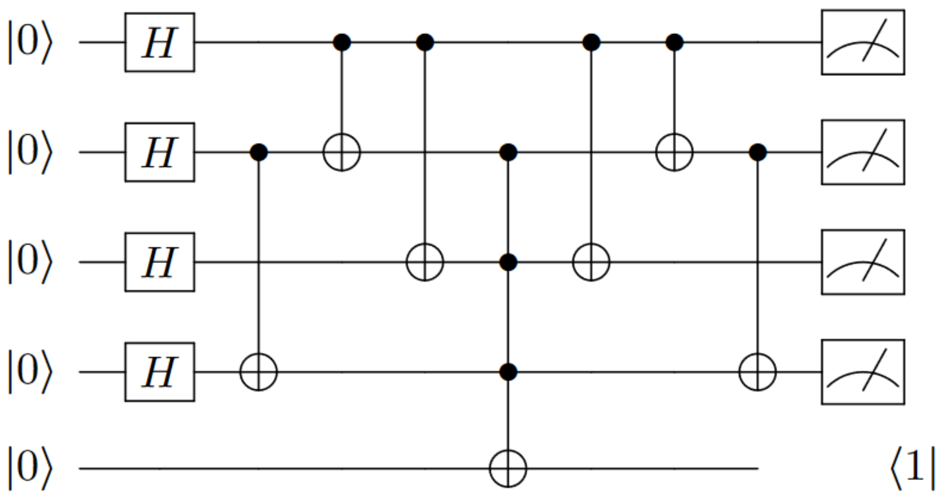
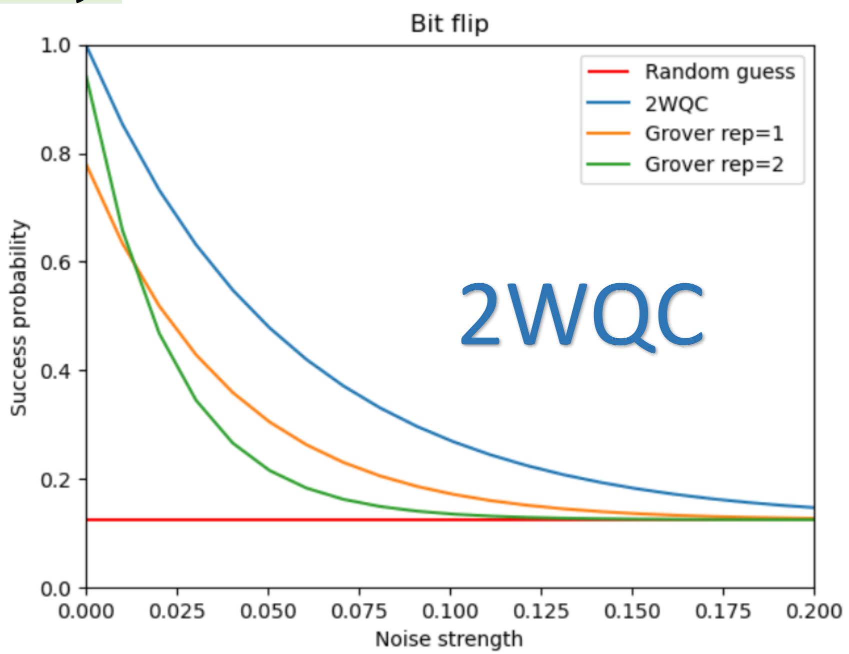
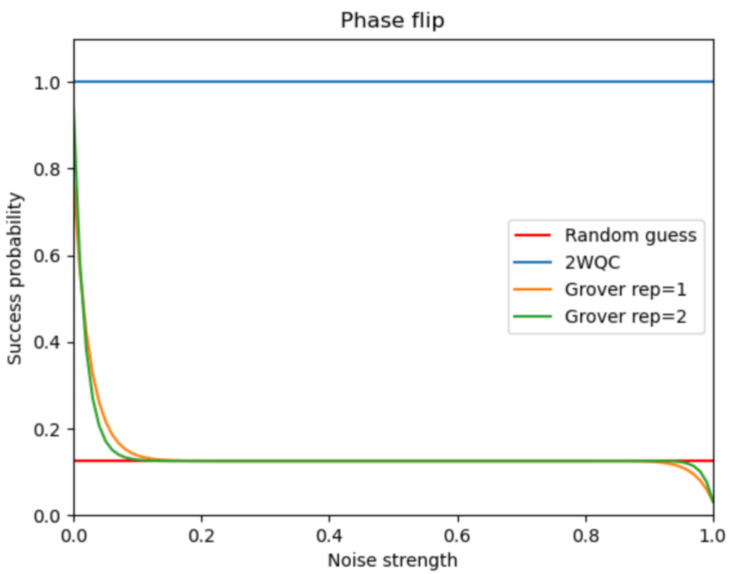
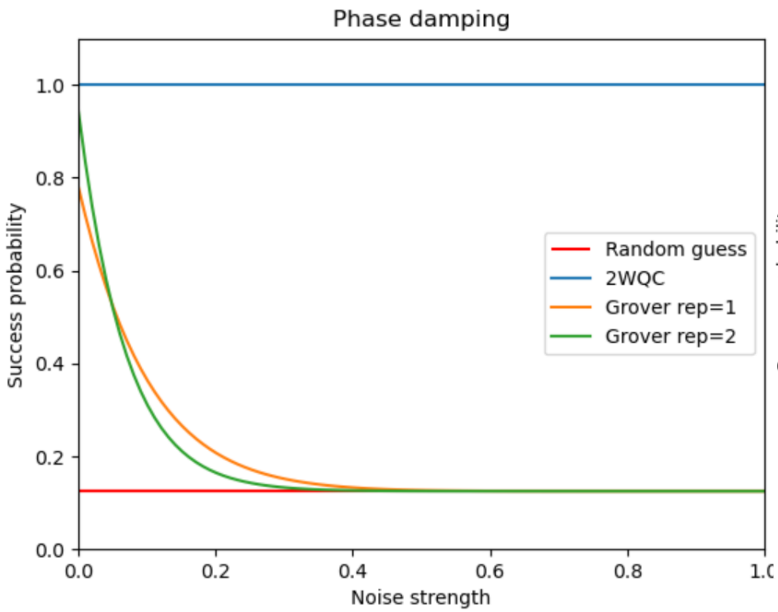
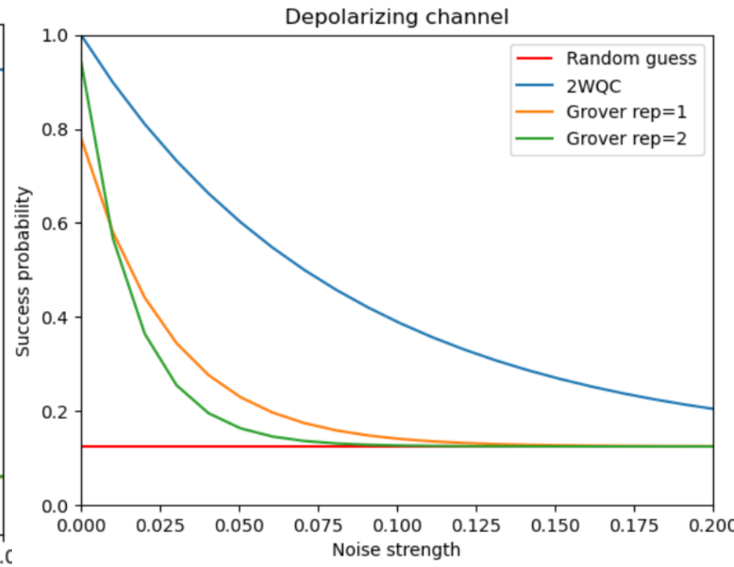


FIG. 6: Quantum circuit for 2WQC Grover solving Sudoku









NP problem: find input satisfying polynomial time verifier

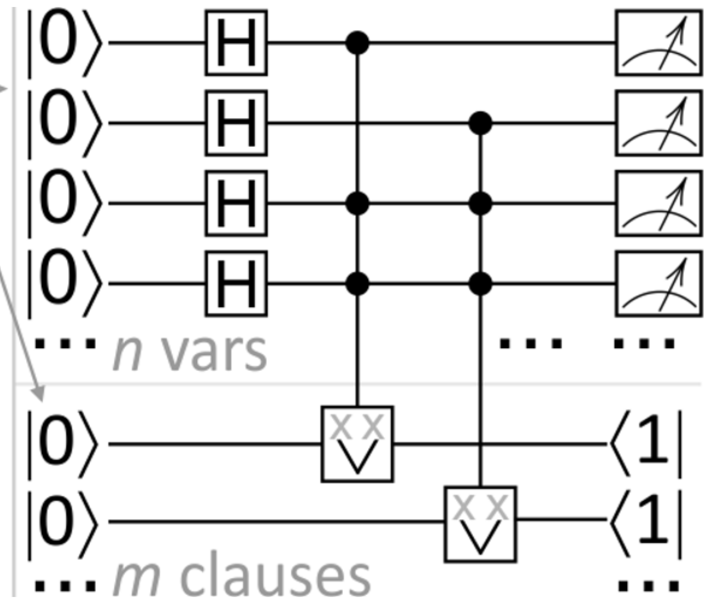
+⟨0| postparation
2WQC in theory
allows NP solvers,
e.g. **cipher breaking**
(resistant PQC???)
global optimizers
like **drug design** ...
Also 2WQC allows
better stability,
error correction

for example 3-SAT problem, like finding $x_1, x_2, \ldots, x_n \in \{0,1\}$:

$$\exists_{x_1 x_2 \ldots} (x_1 \vee \neg x_2 \vee x_3) \wedge (\neg x_4 \vee x_2 \vee \neg x_3) \wedge (x_5 \vee \neg x_4 \vee x_2) \wedge \ldots ?$$

basic 3-SAT setting:

***n* variables** used up to 4 times,
***m* clauses** using 3 variables
- prepare ensemble of 2^n inputs
- calculate C-ORs with NOTs: $\boxed{\vee}$
- enforce all C-ORs to 1 with ⟨1|
- measure inptut qubits



Shor quantum routine, measurement restricts to $\{b: y^b \bmod N = m\}$:

$$|\mathbf{00}\rangle \xrightarrow{H_{\mathrm{I}}^{\otimes n}} \sum_{a=0}^{2^n-1} |a\rangle|\mathbf{0}\rangle \xrightarrow{\text{classic}} \sum_a |a\rangle|y^a \bmod N\rangle \xrightarrow{\text{meas}_{\mathrm{II}}} \sum_b |b\rangle|m\rangle \xrightarrow{\mathrm{QFT}_{\mathrm{I}},\text{meas}_{\mathrm{I}}} |c\rangle|m\rangle$$

3-SAT attack (NP), $\langle \mathbf{1}|_{\mathrm{II}}$ restricts ensemble to $\{b: \mathrm{SAT}(b) = \text{true}\}$

$$|\mathbf{00}\rangle \xrightarrow{H_{\mathrm{I}}^{\otimes n}} \sum_{a=0}^{2^n-1} |a\rangle|\mathbf{0}\rangle \xrightarrow{\mathrm{SAT?}} \sum_a |a\rangle|\mathrm{SAT}(a)\rangle \xrightarrow{\langle \mathbf{1}|_{\mathrm{II}}} \sum_b |b\rangle|\mathbf{1}\rangle \xrightarrow{\text{meas}_{\mathrm{I}}} b$$

for imperfect ⟨**1**| would leave exponential number of false solutions

Post-quantum cryptography (PQC): now focused on Shor, Grover

What if better algorithms, upgrades like 2WQC are there/coming?

NP solver verifier: does decryption with given key lower entropy?

Are some of current PQC already resistant? (NP-hard is not enough)

Building nextgen PQC: immune/resistant to quantum NP solver?

E.g. **require initialization**: large calculations based on cryptographic key before proper decoding (tough for key superposition)

Maybe based on **higher class like PSPACE** (private/public key?)

https://en.wikipedia.org/wiki/PSPACE-complete

e.g. formal languages, **3-SAT + ∀** quantifier $\left(\forall_{x,\ldots} \exists_{y,\ldots} (\vee \vee) \wedge \ldots\right)$,

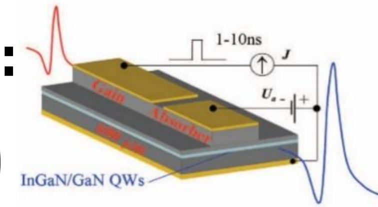
reconfiguration: find path satisfying constraints (~arXiv:1204.5317),

puzzles/games: multiple-interaction cryptography (before low entropy)

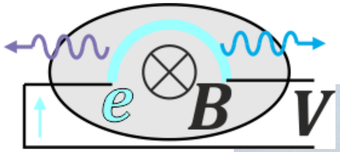
How to generate and measure **negative radiation pressure?**

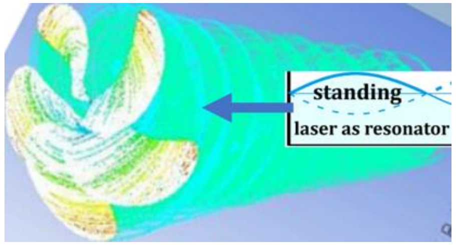
Is CPT symmetry still valid in macroscopic case?

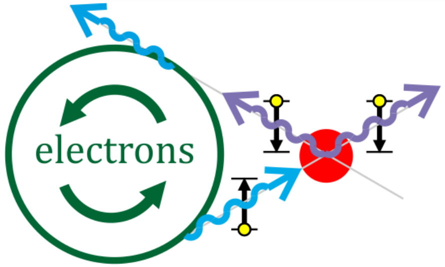
No – we need to modify physics, **Yes – many new applications:**

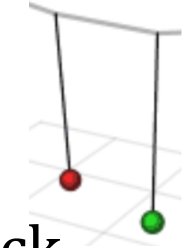


- **Backward beam causing only deexcitation** (transparency?)
  anode tube? antenna? impulse, ring laser? synchrotron/FEL?



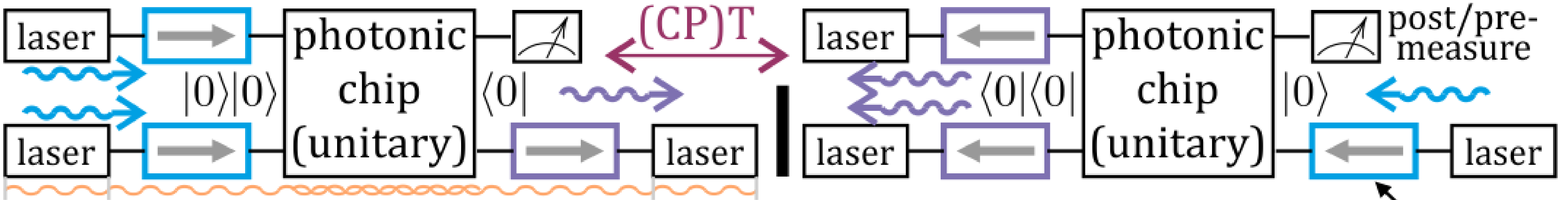






- **Direct Rabi** removing OI, **STED without bleaching** by reversing OI?
- **Causing deexcitation:** radiotherapy, photolithography, nuclear ...
- **Backward camera:** e.g. **CPT(black hole) = white** only emitting?
- **Backward lighted camera** enhancing target deexcitation, BB84 attack,
- **Emission CT**: backward beam + backward camera: medical, geology?
- **2WQC: two-way quantum computer:** solving NP, better error correction

2WQC: adding postparation as state preparation process in CPT perspective



Couple **laser resonators around photonic chip** (directing by optical isolators?)

Potential tests/applications e.g. in synchrotron Solaris?

- **speedup of nuclear deexcitation e.g. Er-169 (forward/backward)?**
- **linac: forward/backward window in new bending magnet?**
- **mirrors to below(/above) for backward lines e.g. in lower floor?**